Controlled Feedback Comparisons and Adaptation Limits in Authenticated Wireless Sensor Routing

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Abstract

Forwarding silence can reflect loss rather than misconduct, but censoring it also prevents missing observations from reversing earlier positive evidence. We examine this trade-off in a separately frozen 1,440-run Cooja factorial study and retain the earlier 750-, 180-, 240- and 600-run experiments. Twelve perturbed geometry blocks combine three retained checkpoints, two radio seeds per block, relay-identity permutations, both attacker positions, benign controls and two timing settings. Four matched arms vary missing-evidence penalties and proof generation while sharing ACK eligibility, expiry and positive-credit handling; an active-proof reference retains its legacy evidence semantics. With scheduled processing, enabling negative updates changes post-onset delivery by +21.170 percentage points (95% pointwise interval +14.410 to +27.834; Holm-adjusted p=0.005859). Enabling proof generation changes it by +1.017 percentage points (95% pointwise interval -0.826 to +2.906; Holm-adjusted p=0.668). Archive diagnostics find 7,023 risk-coordinate differences across 11,927 matched censored–frozen decisions without a relay-choice change; 1,010 of 4,800 offline risk interventions change actions. The O3 and LTO compiler profiles each miss 68 of 200 original signed-receipt deadlines. The contribution is auditable attribution of feedback and traffic interventions within a declared fixture family, rather than a new learning algorithm or a universal switching threshold. Public fixture keys, narrow geometry coverage and illustrative scheduled delays restrict security and deployment inference. Physical RF, measured energy and production device protections remain unvalidated.

Keywords: Wireless sensor networks; Routing feedback; Authentication; Policy transfer; Reproducibility; Virtual validation

1. Introduction

Multihop sensor networks depend on link estimates, progress toward the sink and control traffic. RPL, expected transmission count and the Collection Tree Protocol use these ingredients in different ways [1, 2, 3]. Adding forwarding evidence changes the balance: an acknowledgment can improve confidence in a relay while consuming airtime that would otherwise carry data. Assessing a feedback mechanism therefore requires measuring both the information it provides and the traffic it creates.

Watchdog monitoring and explicit acknowledgments have long been used to detect forwarding misbehavior [4, 5]. Their observations need careful interpretation. Overhearing confirms a transmission attempt, while reception by the intended next hop remains uncertain. Silence is also ambiguous: the first hop may have failed, the channel may be busy, or the observer may have missed a

frame. Penalizing every silence can redirect traffic even when no node is malicious. We study how routing should handle this missing evidence; we do not train an intrusion classifier.

Recent studies combine trust with edge-assisted routing [6], optimizer-based next-hop selection [7], learned trust estimates [8], failure-aware deep Q-learning [9] and opportunistic forwarding [10]. Trust-gated reinforcement learning [11] and secure deep reinforcement learning [12] address related interactions between routing and security. Table 1 compares their stated mechanisms with the question examined here. Differences in channels, workloads and attack models prevent a defensible numerical ranking against those published results.

Table 1. Literature and experimental focus

Study	Verified mechanism	Relation to this experiment
Watchdog [4]	Overheard forwarding attempts	Established passive observation; reception ambiguity remains
2ACK [5]	Reverse acknowledgment scheme	Explicit feedback adds traffic; no numerical cross-study ranking
TARM [6]	Edge assisted trust routing	Trust-aware routing context; different state and deployment
PathGuard [7]	Trust and optimizer next-hop selection	Publisher abstract verified; not independently replicated
SMART [8]	Learned multi-feature trust estimation	Detection task differs from forwarding evidence preference
FRDR [9] and RLFS-OR [10]	Failure-aware and opportunistic learning	Different radio and duty-cycle assumptions
TG-RL-SR [11] and secure DRL [12]	Trust gated and secure routing	Recent context; claims not treated as matched experimental effects
This study	ACK gated passive evidence; four controls	Frozen 30-fixture evaluation with retained negative stages

Earlier architecture comparisons found no joint advantage over a multilayer perceptron of similar capacity. The earlier 600-run telemetry study likewise found no advantage after correction for multiple testing and exposed the traffic cost of active proofs. Both stages and their raw outputs are retained. They motivated the present comparison of receipt-assisted passive feedback with active proofs, subject to prespecified margins against a frozen-prior control on new fixtures.

1.1 Feedback performance and failure boundaries

The benchmark examines when suppressing negative updates from missing evidence helps delivery and when it leaves a routing policy unable to revise earlier favorable evidence. ACK-gated censored passive feedback is compared with active proofs, naive passive feedback, frozen priors and capacity-matched policies. Own first-hop ACK success gates credit; each observer–relay–packet record retains its strongest matched observation; missing evidence leaves the effective failure count unchanged. Packet-reception callbacks connect the ledger to simulated radio events. In Fig. 1, Eq. (1) supplies the candidate shortlist, while Eq. (2) scores the retained candidates. Figure 2 maps the record updates in Eqs. (3)–(4). The benchmark builds on established watchdog, censoring and learning methods.

The original prespecified decision combines a blackhole delivery comparison against active proofs with delivery, critical-service and energy-efficiency margins against frozen-prior attention. Sections 5.10–5.11 add exploratory diagnostics and a separately frozen matched-factorial experiment. The latter isolates missing-evidence penalties and proof generation under common ACK, expiry and late-positive

rules, while counterbalancing relay identity and attacker position. These are controlled comparisons and implementation checks; they do not establish a new learning algorithm, a universal operating boundary or production security.

A separate implementation stage evaluates frozen policies using authenticated messages and node-local transit records in 180 radio runs, followed by serial-interface, firmware-emulation and Cooja checks. Sections 2–4 retain the original feedback benchmark; Section 5 evaluates the changed transport and execution conditions. Section 6 connects the conditional gain and transfer failure without pooling their outcomes or treating their different observation models as an isolated causal contrast.

2. System model and methods

2.1 Radio traffic and threat model

The simulated network has 100 stationary sensors in a 150 × 150 m field and a sink at (75, 75) m. The ns-3.44 low-rate wireless personal area network model provides unslotted access, MAC acknowledgments and packet reception [13]. The log-distance model uses exponent 3 and a reference loss of 40 dB, increased to 54 dB in the weak-link setting. A separate 50 m range limit applies. These settings describe the simulated radio environment.

Runs last 4 s. After a 0.4 s warm-up, the source offers 60 packets at 20 ms intervals, shortened to 2 ms for the congested burst. The three traffic classes have probabilities 0.35, 0.45 and 0.20, with deadlines of 90, 350 and 1,500 ms. Each 64-byte application frame contains 12 identifier bytes and 52 payload bytes; MAC and physical-layer overhead is additional. Nodes broadcast application telemetry every 0.5 s with a 3 ms node offset. We call these messages beacons, although they are distinct from IEEE superframe beacons. Table 2 lists the shared settings.

Table 2. Frozen network settings

Parameter	Value
Sensors and sink	100 sensors; one central sink
Field and range	150 × 150 m; 50 m hard range
Radio engine	ns-3.44; unslotted IEEE 802.15.4
Propagation	Log distance exponent 3; reference loss 40 or 54 dB
Run and offered load	4 s; 60 packets; 0.4 s warm-up
Source intervals	20 ms normal; 2 ms congested
Payload and identifiers	52 and 12 bytes; 64-byte application frame
Telemetry broadcast and cache	0.5 s period; 3 ms/node staggering; 0.75 s expiry
Feedback expiry and poll	150 ms; 1 ms
Traffic classes	Probabilities 0.35, 0.45, 0.20; deadlines 90, 350, 1,500 ms
Blackhole ratio	0 or 25%; non-forging forwarding faults
Assumed voltage and currents	3 V; TX 17.4 mA; RX/listen 18.8 mA; off 20 µA
Policy parameters	Attention 63,368; MLP 63,554; fixed weights
Holdout	30 fixtures × 5 settings × 5 arms = 750 runs

For each blackhole fixture, 25% of sensor identities are selected independently as faulty relays. They accept packets but skip the forwarding decision, while retaining the modeled beacon and receipt behavior. The threat model excludes forged control values, collusion and a compromised sink. Attack

labels are used only by the simulator and evaluator; the router sees forwarding observations, not those labels.

2.2 State construction and frozen routing policy

A received beacon creates or refreshes an observer-specific cache entry containing energy fraction, pending-request pressure, timestamp and link-quality indication. Entries expire after 0.75 s. The current node, visited nodes, exhausted nodes and nodes beyond 50 m are excluded from relay selection. The following fixed score retains at most eight candidates:

$$S_{uv} = 1.25\, g_{uv} + 0.75\, q_{uv} + 0.25\, e_v - 0.55\, R_{uv} - 0.20\, Q_v \tag{1}$$

• For observer u and candidate v, g is sinkward progress divided by 50 m and q is the current estimated MAC success probability; both are dimensionless.

• Candidate energy fraction e is remaining modeled energy divided by 0.5 J; Q is pending-request pressure normalized by 32; R is the dimensionless risk score from Eq. (5). The preselection score S is dimensionless.

The observation contains 13 context values and eight candidate rows of ten values each. Context records local energy and queue state, normalized sink distance, remaining deadline, traffic class and six node-local service-debt coordinates, updated only at sources in this benchmark. A candidate row records whether the node is the sink, its energy fraction, estimated MAC success, received link quality, queue pressure, sinkward progress, evidence risk and uncertainty, expected retries and normalized link delay. During radio transfer, received link-quality indication replaces the RSSI proxy used in training. The weights remain fixed despite this feature shift.

Equation (1) uses hand-set preselection coefficients, not learned parameters. The retained source records the values but contains no systematic tuning history. The later ±20% coefficient experiment in Section 5.5 is a local sensitivity check and does not retrospectively optimize this score. Equation (2) divides the query–key product by the square root of 96, as implemented in PyTorch.

Shared encoders map context x and candidate h to separate 96-dimensional representations. Their query–key interaction and a shared candidate bias give the action logit:

$$\ell_{uv} = \frac{q_\theta(x_u)^T k_\theta(h_{uv})}{\sqrt{96}} + b_\theta(huv) \tag{2}$$

• For context vector x at observer u and candidate feature vector h for link (u,v), ℓ is a dimensionless action logit. Both feature vectors contain normalized, dimensionless coordinates in the frozen benchmark.

• qθ and kθ are learned query and key transforms; bθ is the learned scalar candidate bias. The symbol qθ denotes a neural transform, whereas q in Eqs. (1) and (4) denotes link success. Parameters θ are fixed during evaluation.

A masked argmax chooses the relay. Candidate encoders share their weights, and the critic pools valid candidates by a masked mean, following the set-function motivation in [14]. The attention model has 63,368 parameters; the capacity-matched perceptron has 63,554 parameters and 203 hidden units. Candidates are ranked by the preselection score, with configured node identity breaking ties. Shared scoring therefore does not make the entire routing pipeline invariant to arbitrary node relabeling.

Both model families were trained on seeds 41001–41030 for 400 episodes per seed. Training used eight updates of 50 episodes, each followed by three full-batch PPO epochs. The learning rate was 0.0003, clipping parameter 0.2, discount 0.99, generalized-advantage factor 0.95 and entropy

coefficient 0.01 [15, 16]. Six cost channels represented class-specific delay violations and losses. Lagrange multipliers used an update rate of 0.03 and were clipped to [0, 50]. This finite-budget procedure does not inherit the guarantee of constrained policy optimization [17]. Checkpoints and thresholds were held fixed throughout the new radio experiments.

2.3 Receipt assisted censored evidence

Observer u maintains a separate record for each selected sensor relay v and packet p. Evidence strength s and credited strength c are indexed by that record; effective counts α and β are shared across records for the same observer–relay pair. Each record stores the link-success estimate at registration, ACK status and expiry. A matched overheard forwarding frame contributes strength 0.5 and a reverse receipt generated by sink delivery contributes strength 1.0. Equation (3) retains the maximum, suppressing duplicate credit.

$$s_{uvp} \leftarrow \max\{s_{uvp},\ 0.5\, I_{overheard}, I_{receipt}\} \tag{3}$$

• s is the strongest matched evidence for record (u,v,p), initialized to 0 and bounded by 1; it is dimensionless.

• I_overheard and I_receipt are dimensionless binary indicators of a matched forwarding-frame event and reverse-receipt event for the same record. Their weights are 0.5 and 1 in Eq. (3).

Credit is assigned only when the observer's own first hop is acknowledged and s exceeds c. Evidence received before the ACK is stored. When that ACK arrives, the same eligibility check applies to the stored evidence. In Eq. (4), q denotes the value cached when record (u,v,p) was registered, rather than a later link estimate. The update is executed only for a strictly positive uncredited increment:

$$\alpha_{uv} \leftarrow \alpha_{uv} + \ clip\left(q_{uv} \max\{0, s_{uvp} - c_{uvp}\},\ 0.05,\ 1\right), c_{uvp} \leftarrow s_{uvp} \tag{4}$$

• α is the dimensionless effective success count for observer–relay pair (u,v), initialized to 1; c is the dimensionless credited strength for record (u,v,p), initialized to 0.

• q is the dimensionless cached registration-time link-success estimate. The clip operator limits each positive increment to [0.05, 1]; c is then set to the current s.

In the censored arm, β stays at its initial value of 1 for each observer–relay pair. A failed first hop closes the record without positive credit. For an acknowledged first hop, expiry is set 150 ms after the ACK; absent forwarding evidence causes closure without a failure increment. A later matched observation can still increase s and earn previously unassigned credit. For $q<0.1$, the 0.05 floor can make two half-credit updates exceed one full q-weighted update. The smallest q in the audited decisions was 0.6923, so this case did not occur in those decisions.

The static floor in Eq. (4) acts on effective counts and is not a Bayesian update. Max-strength deduplication bounds the evidence state but does not remove the clipping artifact. Equation (5) uses 1.645 as a fixed multiplier on a beta-form standard deviation; neither this coefficient nor the score clipping provides calibrated 95% coverage or an exact Beta quantile. The fixture-level confidence intervals reported in the results are calculated independently of this routing heuristic. Changing these scoring rules would define a different feedback mechanism.

With the rate-anomaly term disabled, the retained policy maps the pair-specific effective counts to a risk coordinate ρ and an augmented routing score R:

$$\rho_{uv} = 0.75\left(1 - \frac{\alpha_{uv}}{\alpha_{uv}+\beta_{uv}}\right), R_{uv} = clip\left(\rho_{uv} + 1.645 \times 0.75\sigma_{uv},\ 0,\ 1\right) \quad (5)$$

• ρ and R are dimensionless scores for observer–relay pair (u,v). The factor 0.75 scales both the failure-related coordinate and its uncertainty term; clipping bounds R to [0, 1].

• σ = sqrt[αβ / ((α+β)²(α+β+1))] is dimensionless, with α and β taken from the same observer–relay pair. The beta-form expression supplies a score component, not an evaluated posterior interval.

Table 3 sets out the feedback controls. Naive passive uses the same observations and duplicate suppression but penalizes an acknowledged forwarding attempt when evidence is still missing at expiry. Active proofs use separate packets and the earlier timeout convention; their credit rule does not require the passive arm's first-hop ACK gate. Frozen prior keeps the initial risk coordinates without updating them. The capacity-matched MLP uses the censored ledger. Thus, the active-versus-censored comparison changes feedback semantics as well as traffic; it does not isolate airtime alone.

Table 3. Feedback controls

Arm	Evidence and expiry	Policy and interpretation
Active proofs	Separate proofs; legacy timeout and ACK eligibility	Attention; joint feedback and traffic comparator
Naive passive	Overhear/receipt positives; missing ACKed forwarding penalized	Attention; missing-observation control
Frozen prior	Same initial risk coordinates; no evidence updates	Attention; score-update control
Capacity MLP	Censored ledger with same radio observations	63554 parameters; representation control
Censored passive	Own ACK required; max-strength credit; missing evidence censored	63368-parameter attention; proposed replacement

2.4 Source service state and simulator boundaries

For a fixed source and traffic class c, the online service debt is updated once when that source settles a packet. The recurrence follows the virtual-queue form associated with constraint tracking [18]:

$$Z_c \leftarrow \max\{0, Z_c + I_{violation} - b_c\} \quad (6)$$

• Zc is the dimensionless debt for class c at the fixed source; bc is the dimensionless allowed violation fraction per settled class-c packet. The source index and update-event index are suppressed in Eq. (6).

• I_violation is 1 if no receipt is available by service expiry or receipt age exceeds the class deadline, and 0 otherwise. A separate loss-debt channel uses the analogous receipt/expiry outcome. These online indicators differ from the sink-arrival endpoint in Eq. (8).

Only sources update these debts, on receipt or a 1.5 s expiry; intermediate relays retain no complete debt for transit service. Receipt age includes the reverse path, so a timely sink delivery can still produce a late source-observed outcome. The simulator maintains a shared visited set and reverse-path map without serializing their dissemination. Beacon energy and queue values are likewise supplied as metadata keyed to a transmitted identifier, with access gated by a radio-receive callback. Figure 1 marks these simulator-held inputs. Section 5 evaluates a separate authenticated implementation that decodes transmitted state.

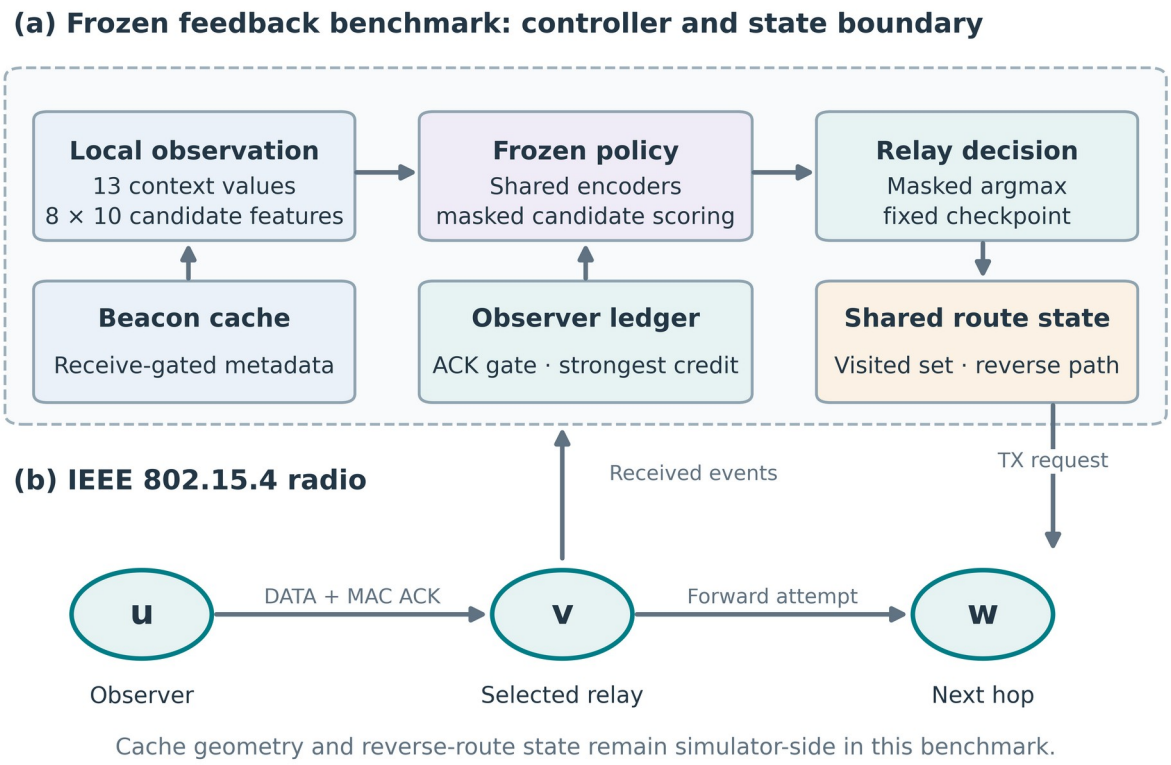


Fig. 1. Frozen benchmark architecture.

Algorithm 1 gives the censored-passive event procedure; Table 3 specifies the other arms. Figure 2 shows the ACK gate and strongest-evidence lifecycle. Radio callback controls check reception of a two-hop forwarding observation, absence of reception at an out-of-range observer and absence of proof frames in the passive arm. They test callback paths rather than authentication or intrusion-classification accuracy.

Algorithm 1: ACK-gated censored passive routing

Input: radio events, cached telemetry, fixed policy θ; censored mode, 150 ms timeout.
Output: next-hop requests, observer evidence and source-service debts.

01. For each received event or 1 ms polling tick:
    02. If a routing request is pending:
        03. Reject visited/exhausted nodes and the 22-hop cap; form at most eight candidates.
        04. If none is feasible, stop this request; otherwise select the masked-policy argmax.
        05. For a non-sink relay, register (u,v,p) and cache q at registration; transmit DATA.
    06. Else if the event is the observer's own first-hop MAC result:
        07. On success, mark ACK eligibility and set expiry to ACK time + 150 ms.
        08. On failure, close the record without incrementing α or β.
    09. Else if matched forwarding evidence or a reverse receipt is received:
        10. Ignore unknown records and closed records without a successful own ACK.
        11. Update the strongest evidence s by Eq. (3), retaining any earlier stronger value.
    12. After either a successful ACK or an accepted evidence event:
        13. If ACK-eligible and s > c, apply Eq. (4); then set credited strength c to s.
    14. At each polling tick, close expired records without a β increment.
        15. Retain acknowledged records so later matched evidence can upgrade s and c.
    16. At a source receipt or 1.5 s service expiry, if the packet is unsettled:
        17. Update its class debt once by Eq. (6), then mark the packet settled.

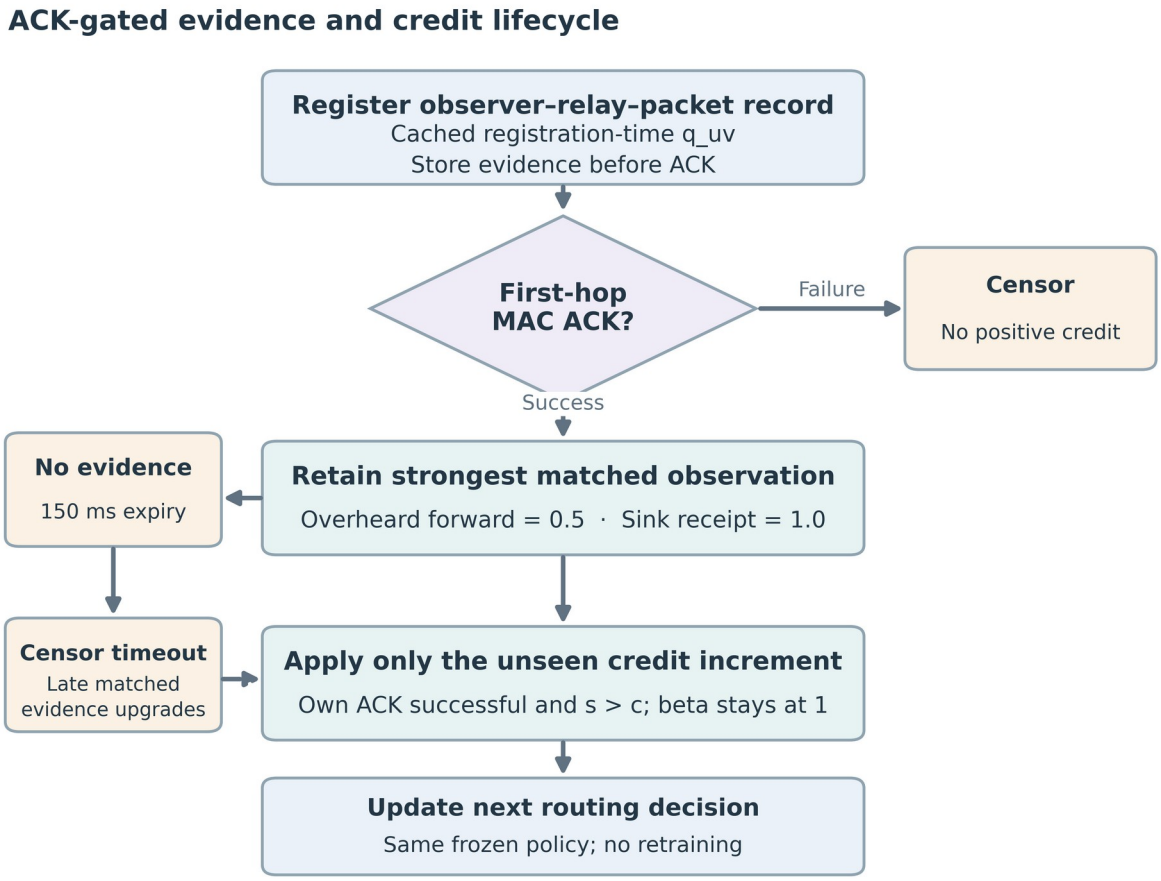


Fig. 2. ACK-gated evidence lifecycle.

3. Experimental design

3.1 Development separation and untouched fixtures

Figure 3 records the source freeze, technical pilot and retained evaluation runs. Telemetry fixtures 76001–76030 informed the redesign. The protocol, source and decision rule were committed as 4027f071322cb5682f39329f656d76df3b68ff62 before fixtures 77001–77030 were evaluated. Technical pilot 76901 is excluded from inference. Each new fixture uses unchanged attention and MLP checkpoints from seed 41001 plus its fixture offset.

The four-second horizon isolates a burst and its feedback. Positions are available at startup and telemetry begins immediately; the model does not simulate RPL or CTP parent discovery. This design therefore evaluates a transient workload rather than routing convergence or steady operation.

Thirty fixtures crossed with five settings and five feedback/policy arms yield 750 runs and 45,000 offered source packets. No run is excluded. Within a fixture, arms share topology, traffic classes, offered packets and blackhole identities. Their event schedules can diverge, so matching seeds does not guarantee identical radio draws at every event. Statistical pairing therefore applies to the complete fixture and checkpoint. Packets are not independent replicates, and the sampled geometries come from one stationary placement family with a central sink.

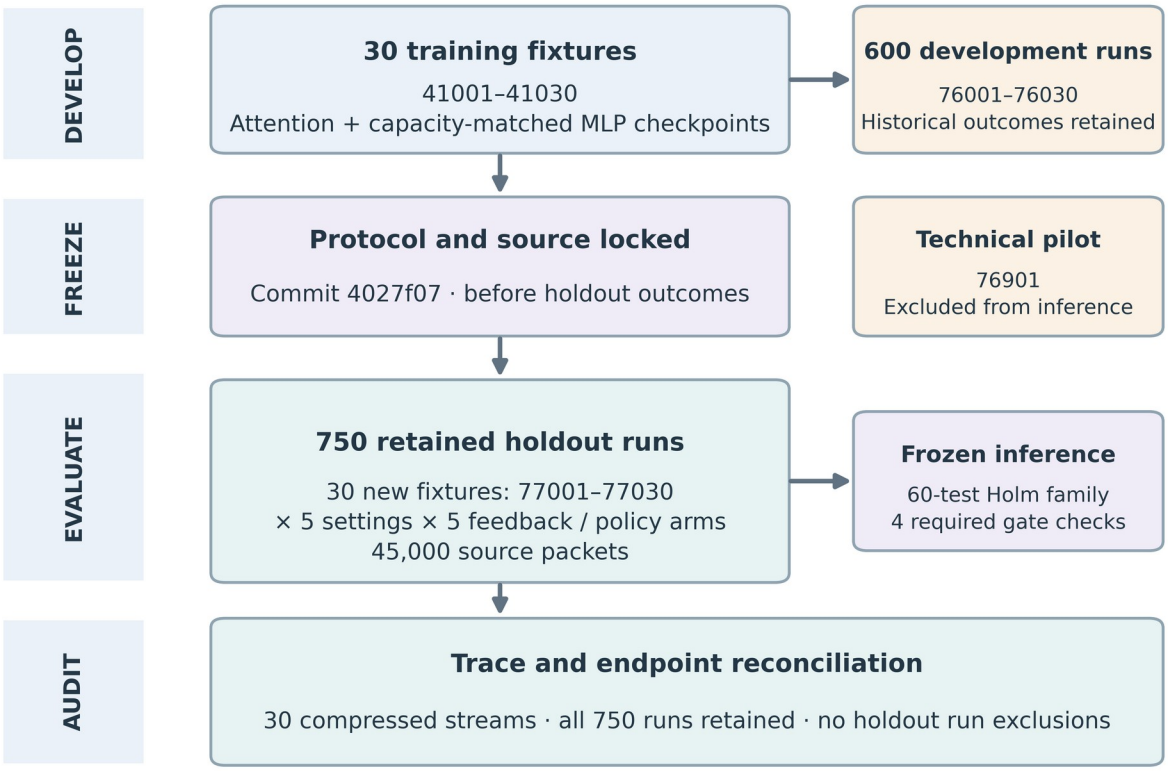


Fig. 3. Frozen evaluation stages.

3.2 Endpoints and radio accounting

Delivery is the fraction of offered source packets received by the sink:

$$D = \frac{N_{delivered}}{60} \tag{7}$$

• D is dimensionless; Ndelivered counts distinct packet identifiers received by the sink during the 4 s horizon.

For critical class 0, a packet violates its requirement if it is lost or arrives after 90 ms:

$$V = \frac{\sum_{p \in C_0} I[p \ missing \ \vee delay_p > 0.09 \ s]}{|C_0|} \tag{8}$$

• V is a dimensionless violation fraction and C0 is the set of offered critical-class packets.

• delayp is sink reception time minus source generation time, in seconds.

PHY-state integration sums the time each sensor spends transmitting, receiving/listening and switched off, including control traffic and retries. Sink energy is recorded separately and omitted from the sensor numerator. With the assumed 3 V supply, modeled radio energy is:

$$E = 3 \sum_{u \in sensors} \left(0.0174\, T_{u,TX} + 0.0188\, T_{u,RX/listen} + 0.000020\, T_{u,off}\right) \quad (9)$$

• E is modeled sensor radio energy in joules; residence times T are in seconds and coefficients are amperes.

• The supply is 3 V. Currents are explicit assumptions, not calibrated measurements; CPU, sensing and cryptography are excluded.

Dividing this expenditure over the common observation window by delivered application payload gives energy per delivered byte:

$$\eta = \frac{E}{52\, N_{delivered}} \quad (10)$$

• η is radio joules per delivered payload byte; 52 is payload bytes per packet.

• A lower η can result from a larger delivered-byte denominator even if total E does not decrease.

3.3 Statistical analysis and frozen gate

For each comparator and endpoint, we calculate one difference per fixture and summarize the 30 paired differences. The two-sided 95% Student t intervals are descriptive and unadjusted. Hypothesis tests retain the prespecified Holm family of 60 comparisons: five settings, four comparators and three endpoints [19]. Delivery and violation differences use percentage points; energy-efficiency differences use J/byte. The comparison family is unchanged after examining the results.

The replacement passes only if all four prespecified checks succeed. Against active proofs, blackhole delivery must improve with Holm $p<0.05$. Against frozen prior, the one-sided 95% bounds must remain above −1 delivery point, below +1 critical-violation point and below an energy-efficiency geometric ratio of 1.02. Table 5 gives the criteria and outcomes. These relative margins assess a feedback replacement; the original absolute class-0 violation budget remains 1.5%.

4. Results

4.1 Delivery and matched controls

Under blackholes, censored passive attention delivered 75.94% of offered packets, compared with 71.28% for active proofs. The paired gain was 4.67 percentage points (95% CI [2.36, 6.97]; Holm $p=0.0161$). Figure 4 compares delivery contrasts across settings; Fig. 5 shows the paired blackhole observations and all control contrasts. Among the delivery contrasts, only the blackhole comparison against active proofs survived the 60-test correction. The energy-per-byte contrast also survives correction (Section 4.2).

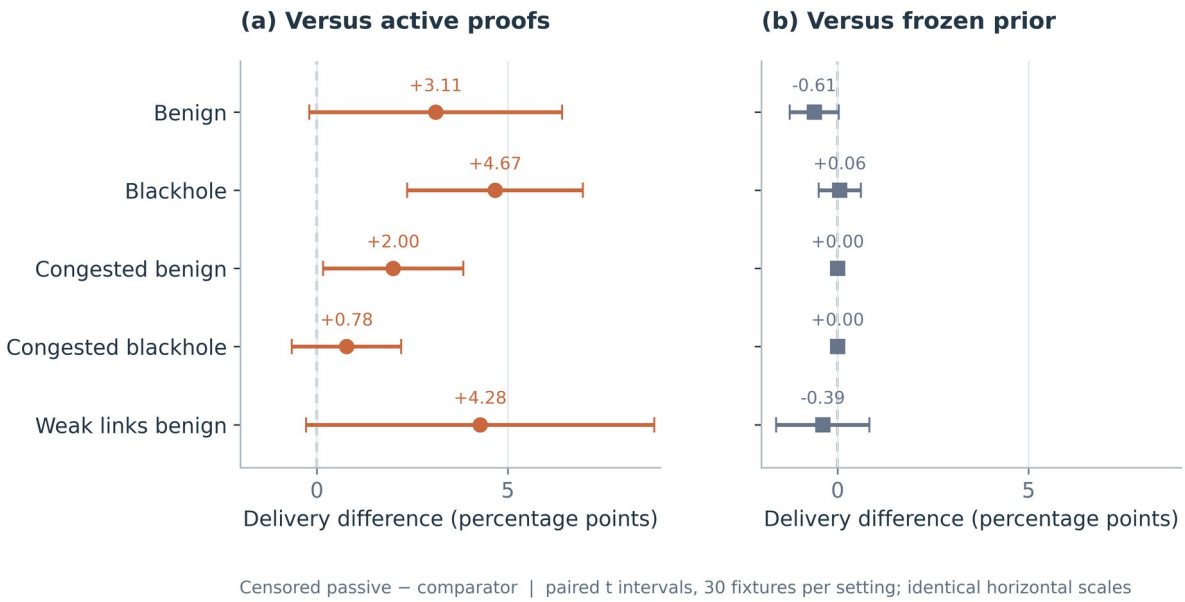


Fig. 4. Delivery effects across settings.

Table 4 reports the remaining blackhole contrasts. Delivery differed by −0.17 points against naive passive, +0.06 points against frozen prior and +0.44 points against the capacity-matched MLP. None was significant after correction. The active-proof comparison consequently does not establish a unique benefit of censoring, evidence updating or attention. The declared margins also do not establish equivalence outside the tested conditions.

Table 4. Blackhole paired effects

Comparator	Delivery points	Critical violation points	Energy mJ per byte
Active proofs	+4.667 [+2.365, +6.969] p=0.01612	-4.518 [-8.253, -0.783] p=1	-0.644 [-0.974, -0.315] p=0.02385
Naive passive	-0.167 [-1.557, +1.223] p=1	-0.017 [-2.381, +2.348] p=1	-0.066 [-0.416, +0.284] p=1
Frozen prior	+0.056 [-0.498, +0.609] p=1	-0.124 [-1.388, +1.139] p=1	-0.011 [-0.078, +0.057] p=1
Capacity MLP	+0.444 [-1.461, +2.350] p=1	-2.384 [-6.065, +1.298] p=1	-0.076 [-0.364, +0.212] p=1

Values are censored passive minus comparator; intervals are unadjusted paired 95% intervals. p values are Holm adjusted over the original 60 tests. Negative critical-violation and energy-per-byte differences favor censored passive. These are differences, not absolute arm means.

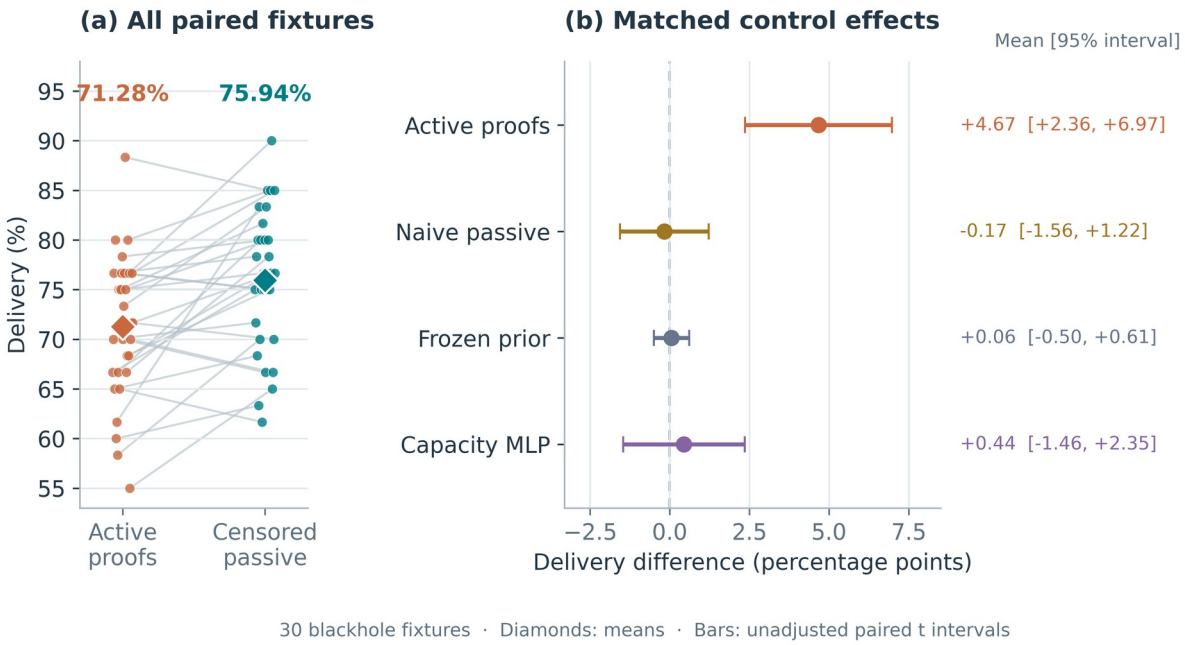


Fig. 5. Blackhole observations and effects.

4.2 Energy efficiency and control traffic

Under blackholes, energy per delivered payload byte decreased by 0.0006443 J/byte relative to active proofs (95% CI [−0.0009740, −0.0003145]; Holm p=0.0239). Total modeled sensor radio energy rose from 22.546943 to 22.547473 J, with a mean increase of 0.000531 J computed before rounding. The lower ratio reflects delivery efficiency rather than reduced total energy. Receiving/listening accounts for an average 99.36% of modeled sensor radio energy across the 750 runs.

Duty cycling changes the cost of passive observation. As an accounting example, an extra 150 ms listening window instead of radio-off time costs 3 × (0.0188 − 0.000020) × 0.15 = 0.008451 J under the assumed currents. Overlapping windows and existing awake periods reduce the incremental cost. This calculation is not a duty-cycled MAC experiment. Such a comparison must charge channel checks, rendezvous, observation windows and proof retransmissions to all arms; the always-listening results do not predict which mechanism would use less energy.

Figure 6 separates total energy, energy per delivered byte and PHY transmission count. Censored passive generates an average 60.57 fewer PHY transmissions per blackhole run than active proofs. All passive and frozen-prior arms request zero forwarding proofs, compared with 3,369 requests across the active arm's 150 runs. Beacons and reverse receipts remain in every arm. In the assumed always-listening model, receive/listen current exceeds transmit current; reducing transmissions therefore need not reduce energy over a fixed window. More delivered bytes explain the lower energy-per-byte ratio.

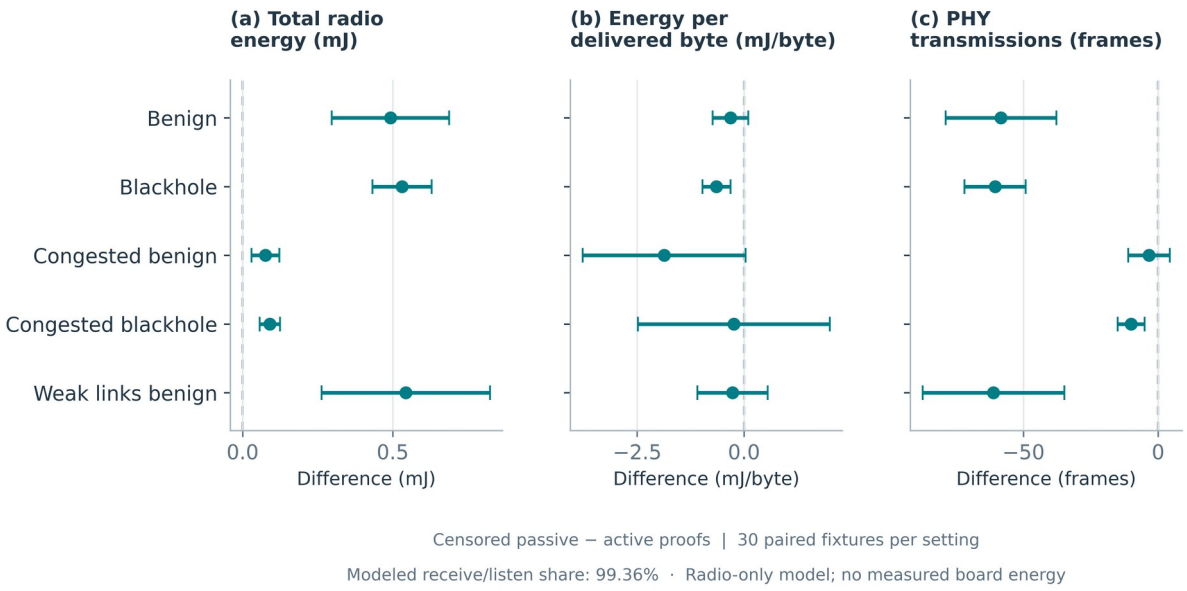


Fig. 6. Modeled radio-energy attribution.

4.3 Gate outcome and service limits

All three blackhole margins against frozen prior passed. The one-sided delivery lower bound was −0.405 percentage points, the critical-violation upper bound was +0.925 points, and the geometric energy-efficiency ratio upper bound was 1.00505. With the corrected delivery comparison against active proofs, these results satisfy all four checks in Table 5. The critical-violation bound lies close to its +1-point margin, leaving little room beyond the prespecified threshold.

Table 5. Prespecified incremental gate

Check	Frozen criterion	Observed result
Delivery versus active proofs	Positive gain and Holm p<0.05	+4.667 points; p=0.0161; pass
Delivery versus frozen prior	One-sided lower bound >−1 point	−0.405 points; pass
Critical violations versus frozen prior	One-sided upper bound <+1 point	+0.925 points; pass
Energy efficiency versus frozen prior	One-sided geometric upper ratio <1.02	1.00505; pass
Joint incremental decision	All four checks required	Pass; absolute QoS compliance not established

The reduction in critical violations against active proofs did not survive correction for multiple testing. Censored passive still violates approximately 26.2% of critical requirements under blackholes, well above the original 1.5% budget. Passing the relative replacement test therefore does not establish absolute deadline compliance or a guaranteed quality-of-service controller.

4.4 Implementation and trace audit

The action and ledger audit checked 67,522 decisions, 24,747 passive observations, 28,353 evidence updates and 75,750 node-energy identities. A separate endpoint audit reconstructed delivery and critical-violation values for all 750 runs from source decision times and sink receptions, then checked beacon counts, decision counts and PHY-state energy. All 30 gzip streams were read to completion. The original-stage suite had 51 passing tests; the pre-extension implementation had 83 passing tests, including the radio callback checks. The post-freeze suite in Sections 5.6–5.7 has 107 passing tests.

Release checks found a truncated trace for fixture 77004. It was regenerated with the same frozen implementation, and all 750 canonical summary rows matched the saved pre-recovery summary exactly. Parameters, checkpoints and analysis rules were unchanged. The recovery record and audit code accompany the raw traces. These checks reconcile the implementation's own records; they do not provide an independent radio model, external packet capture or replay of every beacon-reception and service-state transition.

5. Authenticated transport and virtual validation

5.1 Local state and authenticated evidence

The authenticated branch uses the message path in Fig. 7. Coordinates, modeled remaining radio energy, queue state and transmit time are serialized in 58-byte application telemetry frames. A node relies on its provisioned own and sink coordinates, authenticated received cache, local MAC outcomes and packet history. Pairwise HMAC-SHA256 protects each hop with a 128-bit tag [20]. Provisioned 64-bit boot sessions and a 64-entry replay window reject stale sessions and duplicates. A source authenticator binds the immutable data fields. Ed25519 signs the source, session and packet identity at the sink [21], and the receipt follows node-local reverse records. Application frames are at most 114 bytes.

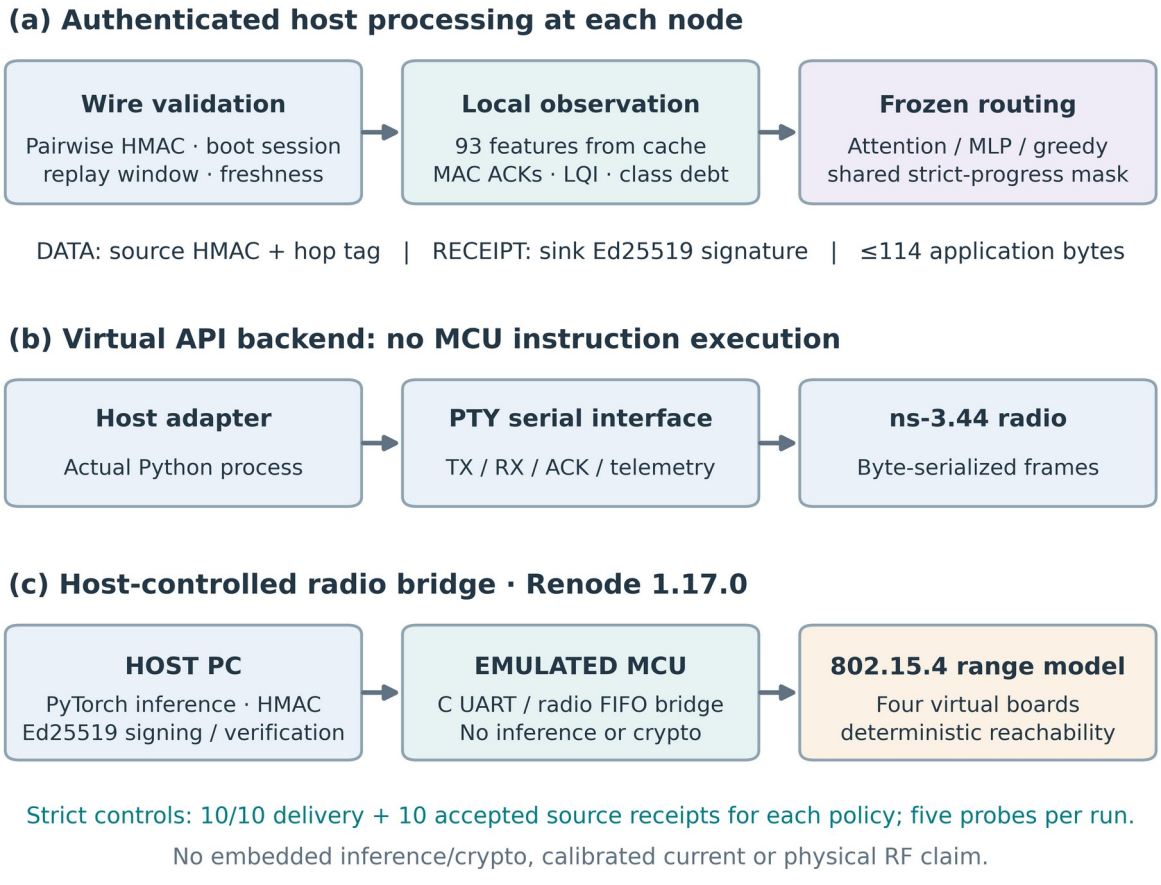


Fig. 7. Authenticated state and virtual interfaces.

Pairwise keys cannot authenticate a transmission between two other nodes, so the authenticated branch in Fig. 7 assigns no credit to third-party overhearing. Unlike Eq. (3), positive forwarding evidence requires a verified sink receipt and a successful local MAC acknowledgment; missing receipts remain censored. The frozen policies therefore encounter a changed observation model rather than an exact

replication of the original ledger. Each node retains at most 128 records, with a 2 s expiry. Keys and sessions are provisioned out of band. Protected key storage, online key exchange and verification of telemetry truthfulness after key compromise remain outside the implementation.

5.2 Frozen host policy transport protocol

The authenticated-transfer protocol and source were committed as 5f06b9213e73206aad11ec8f3a96079a17011a20 before its outcomes were observed. Seeds 83001–83030 define 30 independent topology fixtures. Each pairs a frozen attention checkpoint with its capacity-matched MLP and a greedy progress control. The six-node diamond has two first-hop choices and two downstream relays. Nominal positions are (0,0), (25,−15), (25,15), (50,−15), (50,15) and (75,0) m, independently jittered by ±2 m; range is 38 m. Benign and selective-dropping conditions give 180 runs. In the latter, relay 1 discards even-identifier data after MAC reception. The router cannot access this attack label.

Each run lasts 7.5 s and offers 60 packets carrying 16 payload bytes. Traffic starts at 0.5 s, uses 75 ms intervals and cycles through the three deadline classes. Authenticated telemetry is attempted to every provisioned peer every 0.5 s, staggered by 2 ms. The record includes out-of-range requests and their modeled radio costs. All arms enforce strict progress toward the sink and retain at most eight candidates. Greedy picks the shortest remaining sink distance; the learned policies retain their trained weights. Fixed ns-3 seeds and streams pair the fixtures, although different decisions can consume different random sequences.

The observation has 93 elements: 13 context values and eight ten-value candidate rows. Decoded telemetry supplies neighbor geometry and energy. MAC success starts from a Beta(9,1) prior, and an exponentially weighted ACK-delay estimate uses coefficient 0.2 with an initial value of 5 ms. Received LQI replaces the training RSSI proxy. Queue features use raw MAC pending-request counts instead of normalized training pressure. Forwarding risk uses a positive-only transform of signed-receipt success counts and current link reliability. Nodes maintain separate class-delay and loss debts. These substitutions change the observation distribution; the transfer neither recreates the training model nor calibrates intrusion probabilities.

5.3 Transfer results and audit

Table 6 and Fig. 8 summarize all 180 runs. In benign conditions, attention delivers 81.61% of packets, versus 82.72% for greedy and 83.72% for MLP. Its paired differences are −1.11 points against greedy (95% interval −3.39 to 0.83) and −2.11 points against MLP (−4.50 to 0.28). Under selective dropping, delivery is 61.11% for attention, 62.89% for greedy and 60.50% for MLP. The attention differences are −1.78 points (−6.50 to 2.67) and +0.61 points (−2.56 to 3.78), respectively. Intervals use 10,000 paired percentile-bootstrap samples across fixtures with seed 94001. They are exploratory, without multiplicity-adjusted superiority testing.

Table 6. Authenticated policy transfer.

Condition / policy	Delivery (%)	Receipts (%)	Critical failure (%)	Radio energy (J)
Benign / greedy	82.72	79.22	31.17	2.5188
Benign / mlp	83.72	79.72	30.83	2.5189
Benign / attention	81.61	77.61	31.17	2.5189
Dropping / greedy	62.89	60.56	46.67	2.5204
Dropping / mlp	60.50	58.22	48.50	2.5206
Dropping / attention	61.11	58.17	47.83	2.5205

Means over 30 fixtures per condition and policy. Receipt rates count source-verified sink receipts. Critical failure includes undelivered critical packets. Energy covers modeled radio residence, not board or host CPU energy.

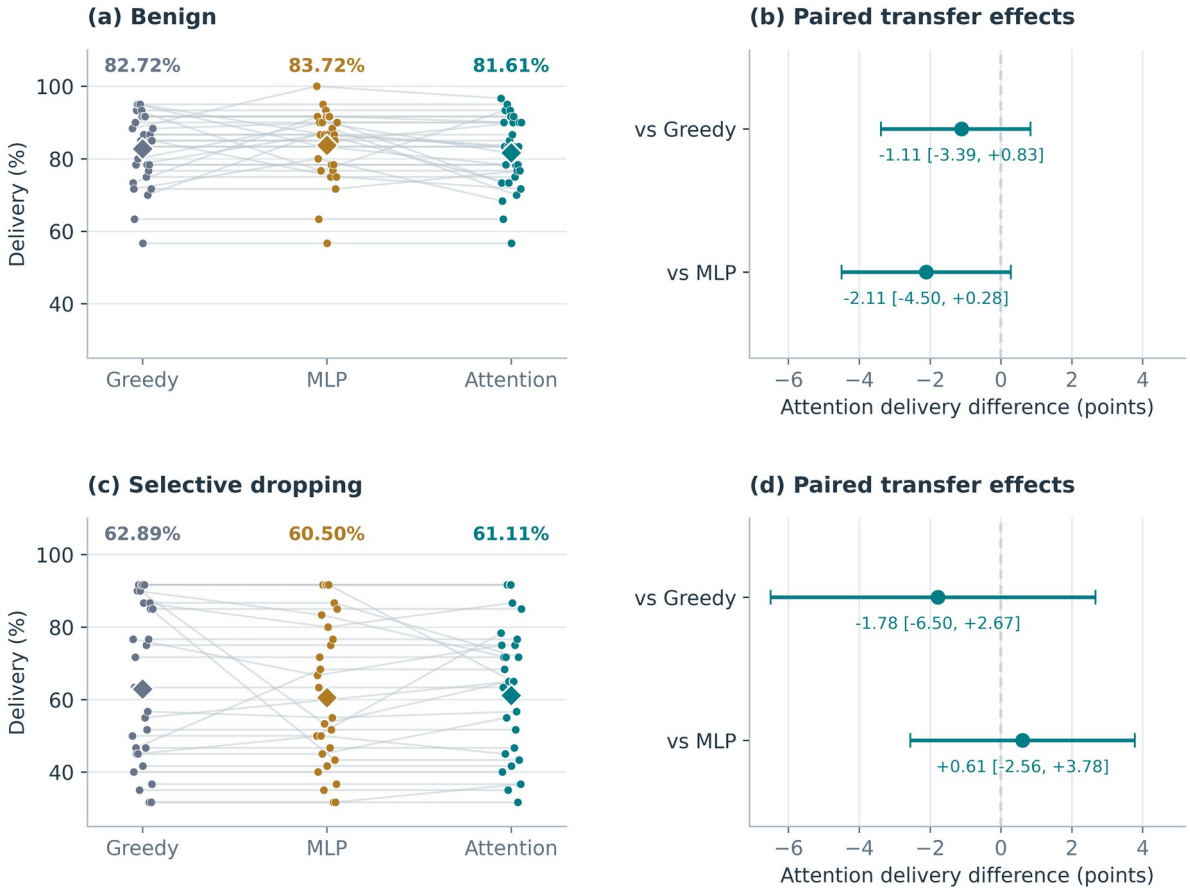


Fig. 8. Frozen policy transfer and paired effects.

The −1.78-point attention–greedy difference under selective dropping has an interval that crosses zero, establishing neither superiority nor equivalence. Replacing the training RSSI proxy with LQI, using raw queue counts and restricting forwarding credit to signed receipts can change the input distribution; their causal effects were not isolated. Greedy avoids neural weights and tensor inference, making it a sensible starting point for deployment assessment, although neighbor state, authentication and packet processing still incur costs.

The frozen host policies operate on serialized simulated-radio frames with multiple relay choices. For attention, critical-deadline failure is 31.17% in benign runs and 47.83% under selective dropping when undelivered packets count as failures. Total modeled radio energy is approximately 2.519 and 2.520 J per six-node run. Always-on listening and repeated authenticated telemetry dominate this radio-only account; embedded CPU and cryptographic energy are excluded.

The audit reconstructed 27,240 decisions, including 14,001 with more than one candidate. It matched 88,538 received byte frames to earlier transmissions, verified 46,204 public sink-signature occurrences and reconciled 1,080 radio-energy identities. Candidate cache values were checked against received beacon bytes, and frozen-policy probabilities were recomputed. Runtime nodes check the secret hop

tags; the private keys are not exported for public verification. This branch leaves the original 750-run traces, checkpoints and decision rule unchanged.

5.4 Host controlled radio bridge firmware

Four pseudo-terminal devices expose the TX, RX, MAC-confirm and telemetry interface while ns-3 carries the application bytes. Four host-adapter processes use the same serial contract intended for later board integration. The first diagnostic delivered five of ten packets and revealed a timestamp error: the host sampled its clock before a blocking read, then used that earlier time to judge freshness. Sampling again when bytes arrived removed the false rejections. The corrected fixture delivers ten of ten packets and settles service at both relays. Both captures are retained. One functional fixture is insufficient to estimate reliability.

The PTY backend models the firmware interface without executing MCU instructions. Its greedy, MLP and attention rechecks each deliver ten of ten packets. A separate Renode 1.17.0 backend runs compiled C firmware on four emulated CC2538 Cortex-M3 boards (Fig. 7). Firmware parses UART JSON, bounds frame sizes, handles the IEEE 802.15.4 radio FIFO and correlates MAC acknowledgments. Routing inference and cryptography remain on the host. Each control delivers ten of ten packets and returns ten verified source receipts.

Five probes check the uncalibrated telemetry label, exact 114-byte delivery with ACK, rejection of 115-byte frames, malformed-hex rejection and failed out-of-range transmission. A separate capture audit checks MCU UART bytes, public sink signatures and packet identities. Development captures retain the startup and metadata corrections. The stricter audit also rejects an older greedy capture with a missing relay-1 final log, despite its saved summary. New runs pass only after all ten offered identities reach the sink, ten signed receipts are accepted at the source, source and relay final records are complete, all five probes succeed and the capture audit passes. All three fresh controls meet these requirements.

The full attention actor–critic stores 63,368 FP32 parameters, or 253,472 bytes (247.53 KiB). That count includes critic parameters that an actor-only implementation could omit. For comparison, the CC2538SF53 variant provides 512 KB flash and 32 KB RAM [22]. Flash-resident weights need not all occupy SRAM, but flash capacity alone says nothing about activation buffers, stack and protocol memory. An embedded build, linker map and peak-memory measurement are needed before claiming feasibility. The host-controlled firmware in this fixture handles UART parsing, frame bounds, radio FIFO access and MAC acknowledgments; it executes neither neural inference nor HMAC/Ed25519.

Renode's deterministic range medium differs from the ns-3 physical-layer model used for statistical evaluation. Energy telemetry is explicitly labeled emulated_no_power_model: the fraction is fixed, while queue telemetry counts pending MCU ACK records. It provides no battery measurement. The chain offers one feasible next hop at each relay, so it tests integration rather than policy selection or reliability. Deadline debts remain nonzero. The emulation does not validate physical clock and pin initialization, physical RF behavior, calibrated current consumption. Section 5.8 adds a separate live MCU policy/crypto pilot; this earlier host-driven fixture is unchanged.

5.5 Post review stress and sensitivity diagnostics

Table 7 and Fig. 9 report a separate exploratory extension on fixtures 84001–84003, using attention checkpoints 41001–41003. Active proofs, censored passive and frozen prior each run for 60 and 300 s, giving 18 runs. Packets arrive every 0.2 s from 0.4 s until near the end of the run: 290 and 1,490 offered packets, respectively. The topology, blackhole fraction, beacon schedule and 0.5 J routing-

energy scale follow the frozen benchmark. These diagnostics were designed after review and do not enlarge the original confirmatory test family.

The extension exposes a limit rather than steady operation. At the assumed listening current, 0.5 J corresponds to roughly 8.87 s. Routing excludes an energy-depleted node, but the radio bridge continues telemetry and PHY-state integration. Delivery consequently plateaus while offered traffic and source-service debt continue to grow. Accumulated radio energy after depletion is bookkeeping for an energized simulation, not energy physically supplied by the original battery. Retained-watch counts also persist because closed records are not reclaimed.

Table 7. Long horizon stress diagnostics.

Horizon / arm	Packets	Delivery (%)	Critical failure (%)	Max. source debt
60 s / Active proofs	290	13.45	85.13	3.94
60 s / Censored passive	290	13.10	85.12	3.94
60 s / Frozen prior	290	12.99	85.43	3.94
300 s / Active proofs	1490	2.62	97.22	15.36
300 s / Censored passive	1490	2.55	97.22	15.36
300 s / Frozen prior	1490	2.53	97.29	15.36

Delivery and critical-failure entries are means across three fixtures; debt is the maximum final delay debt across sources and fixtures. Zero handle-exhaustion events were recorded. Pending-map samples are taken once per second and cannot rule out transient MAC queue drops. This small extension supplies no bounded-queue or stability guarantee.

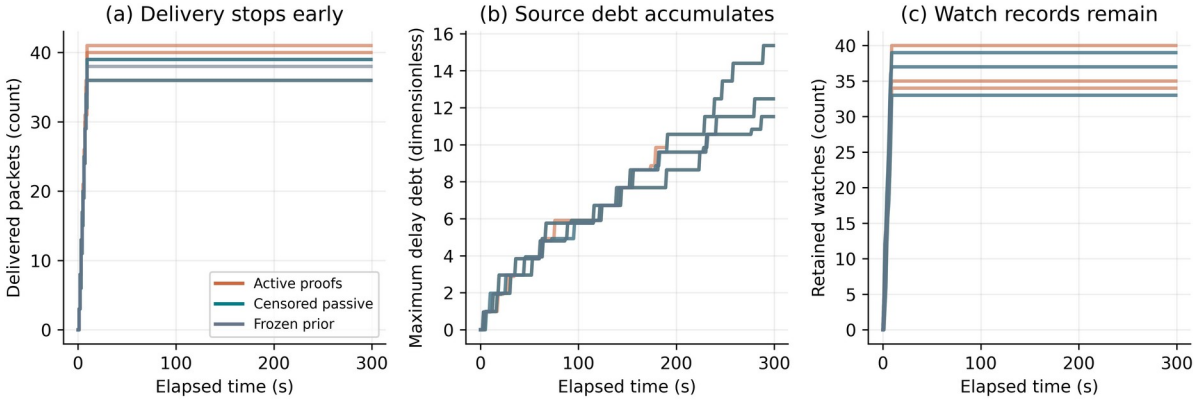


Fig. 9. Executed 300 s stress trajectories.

Figure 10 examines Eq. (1) on three further fixtures, 85001–85003, paired with checkpoints 41001–41003. Each coefficient is changed alone by −20% or +20%; the unchanged setting and ten perturbations give 33 four-second blackhole runs. All 30 perturbed fixture-level delivery contrasts are exactly zero in these runs. This limited result does not establish insensitivity in denser candidate sets or outside the tested range. These are local sensitivity observations, not a tuning procedure or a confidence interval. No checkpoint is retrained and no setting is selected from these results.

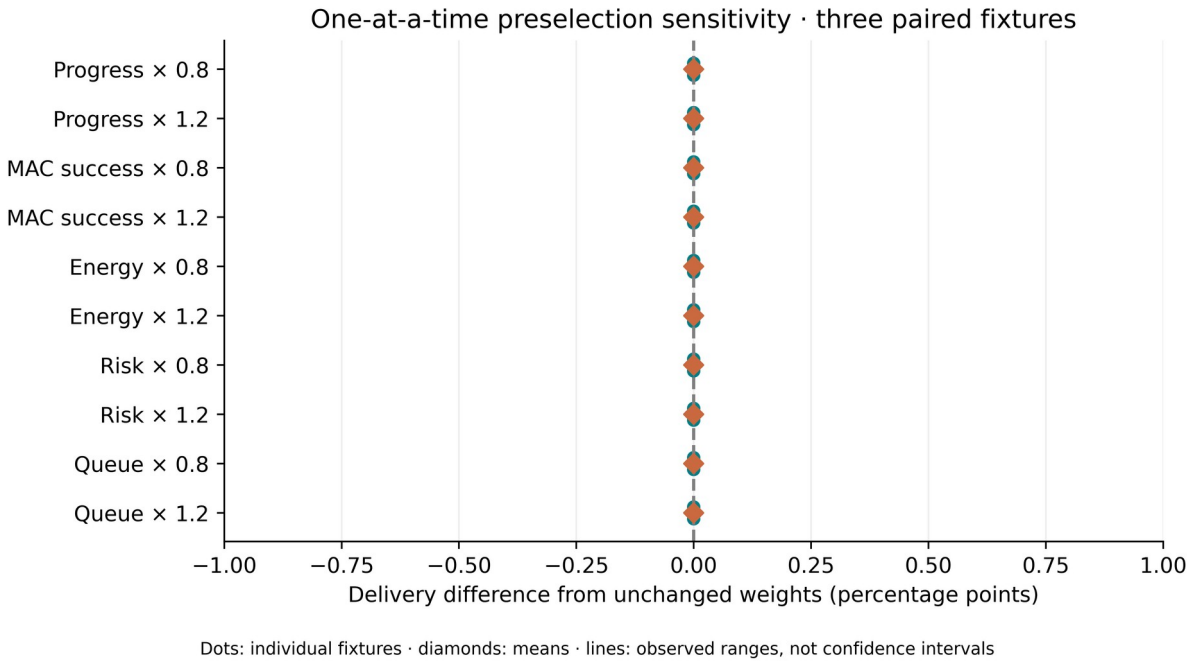


Fig. 10. Executed coefficient sensitivity contrasts.

The inference actor contains 44,161 parameters, requiring 176,644 FP32 weight bytes before runtime buffers; the full actor–critic contains 63,368 parameters. Removing training-only heads therefore changes the storage estimate, but neither count establishes target feasibility. Section 5.8 tests integrated target inference in emulation. Duty-cycled MAC performance, physical drift tolerance and measured energy still require separate validation.

5.6 Factorial traffic controls and power gating

A separate ns-3.47 extension tests proof traffic while holding the evidence rule fixed within each contrast. Its amended source and 30-test analysis plan were committed as 752c11e7fa3c6af427ffe4edb8fa37d3fbcd5bc7 before fixtures 89001–89030 were evaluated. An initial proof-callback error stopped a preliminary 88001 batch; those partial captures and technical pilot 76901 are excluded. Thirty fixtures, four arms and two duty settings yield 240 four-second runs, each offering 60 packets. All arms use the same bounded positive-credit rule, ACK gate, 128-record global simulation-ledger capacity, 2 s retention and frozen attention actor. Unlike Eq. (4), each record clips q once to [0.05, 1] and weights each eligible uncredited increment in s by that fixed value. Proof frames consume traffic but never enter the ledger. Censored versus naive mode changes expiry penalties and late-evidence handling. Table 8 reports arm means; Fig. 11 shows the proof-traffic contrasts.

With always-on listening, adding proofs reduces censored-mode delivery by 6.11 percentage points (unadjusted paired 95% interval −8.94 to −3.28; Holm p=0.00349), and naive-mode delivery by 5.61 points (−7.94 to −3.29; Holm p=0.000852). No evidence-mode or interaction contrast survives the separate 30-test correction. These results identify a traffic contribution in this extension, not the complete cause of the original 4.67-point gain. The nominal 20% awake schedule rejects requests while asleep and has no rendezvous protocol; delivery falls to approximately 13–14%. Busy MAC transactions defer sleep, so nominal duty is not actual PHY residence.

Table 8. Factorial arm means.

Nominal awake	Evidence	Proofs	Delivery (%)	Critical failure (%)	Radio energy (J)
100%	Censored	Off	74.94	24.37	22.547516
100%	Censored	On	68.83	32.51	22.546980
100%	Naive	Off	75.56	24.14	22.547497
100%	Naive	On	69.94	30.89	22.547013
20%	Censored	Off	13.50	87.84	9.791828
20%	Censored	On	13.83	87.71	9.749594
20%	Naive	Off	13.50	87.84	9.804775
20%	Naive	On	13.83	87.71	9.760433

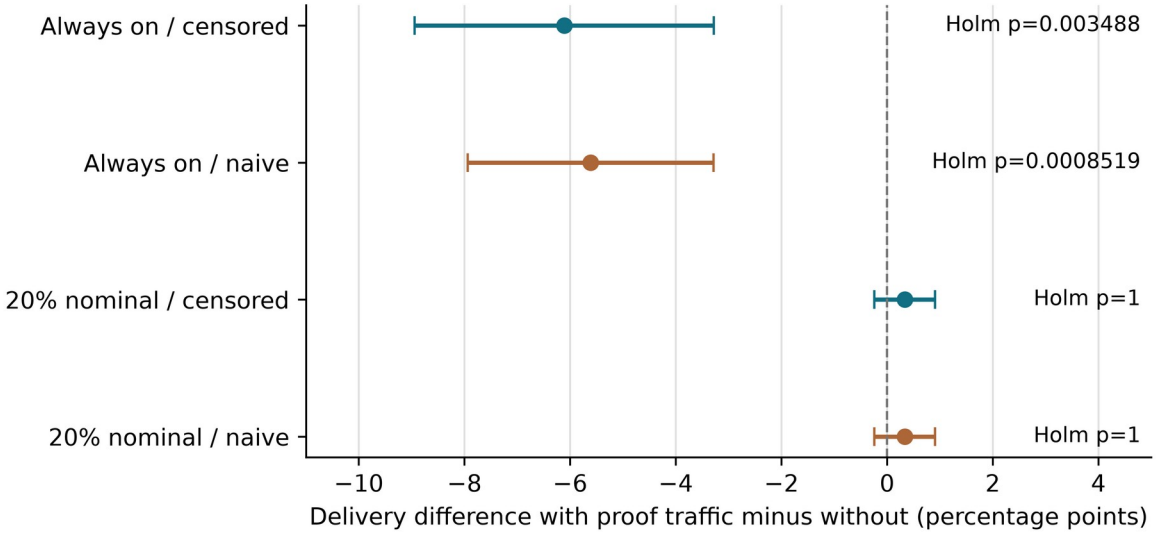


Fig. 11. Proof-traffic delivery effects.

Hard power gating stops the experimental bridge at each sensor's 0.5 J budget. Twenty-seven further 60 s runs cross three fixtures, three feedback arms and three nominal awake fractions. All remain within the 50 J aggregate sensor budget, with zero post-cut transmissions; the audit reconciles 2,727 radio-energy identities. The bounded ledger has zero final watches in all nine runs, whereas legacy controls retain records. All sensors still exhaust their budgets within 60 s. These tests address power accounting and ledger retirement, not sustained service or a whole-system memory bound.

5.7 Clock envelope and Cortex M3 execution

Optional freshness handling uses an externally justified pairwise clock bound. A 40 ms offset bound plus 100 ppm relative drift over 300 s gives 70 ms tolerance. All 675 tested offset/drift combinations accept authentic fresh beacons, while replay and forged-tag checks remain effective. The wider window can also admit older messages near the expiry boundary. It neither synchronizes clocks nor verifies their physical drift; the original default behavior remains unchanged.

A bare-metal C harness executes the 44,161-parameter actor, HMAC-SHA256 and Monocypher 4.0.3 Ed25519 on QEMU 11.1.2's MPS2 AN385 Cortex-M3 model [23, 24]. Eight retained radio decisions and eight candidate-permutation controls produce 128 logits; all 16 actions match PyTorch and span eight candidate indices. The largest logit difference is 1.7881393432617188e-7, below the 1e-5

tolerance. HMAC matches RFC 4231 case 1; Ed25519 signatures match independent Python verification, and a corrupted signature is rejected. Table 9 records the execution checks.

Table 9. Isolated target checks.

Check	Observed result
Actor outputs	16/16 actions; 128 logits
Maximum logit error	1.7881393432617188e-7
HMAC / Ed25519	Known-answer / sign and verify passed
Text and read-only data	341,600 bytes
Observed stack watermark	3,080 bytes
Linker limits	512 KiB flash; 32 KiB SRAM

The image omits radio buffers, RTOS state and integrated protocol memory; its observed stack watermark is not a worst-case bound. Public test keys are used only for validation. This isolated instruction harness and the CC2538 UART/radio bridge in Section 5.4 are distinct experiments. Neither their separate success nor the activation approximation’s agreement on these vectors validates a complete embedded routing system, physical timing or measured energy. Section 5.8 subsequently evaluates a combined radio and policy/crypto image under a narrower four-node protocol.

5.8 Live MCU routing and authentication

The four-node CC2538 pilot executes observation construction, frozen attention inference, hop authentication and receipt handling inside the MCU. The host triggers traffic and beacon exchange and captures UART output. Earlier public fixtures deliver twenty packets and verified receipts, with fifty matching routing decisions. Table 10 compares the retained private-key FP32 streams with two later implementations; each profile offers one hundred packets in a three-hop chain and one hundred in a geometry with two feasible source choices. These are sequential functional diagnostics, not independent statistical replicates or a new confirmatory comparison; N is the offered count.

Operation-level profiling identifies approximately 270 ms per-hop FP32 inference and 81 ms Ed25519 verification in the declared emulator. A separate Q22 implementation uses signed 32-bit weights and activations, 64-bit accumulators and an 8193-point tanh lookup table. Export checks bound arithmetic for input coordinates in [−32, 32]; nonfinite or out-of-domain inputs are rejected. The lookup table adds 32,772 bytes beyond the 176,644 parameter bytes. All five hundred retained decisions and the later MCU decisions match the FP32 reference at their recorded observations, with logit errors below 1e-4. This is tested approximation agreement, not universal action equivalence. Inference falls to approximately 12–13 ms with one candidate. Caching the provisioned sink signing key also removes repeated key derivation from the packet path.

Table 10. MCU receipt profiles.

Profile / topology	N	Delivered / Ed proofs	Ed delay (ms)	Ed misses	Fast delay (ms)	Fast misses
FP32 / chain	100	100 / 100	1134–1142	67	—	—
FP32 / choice	100	100 / 100	925–931	67	—	—
Q22 / chain	100	100 / 100	331–339	34	—	—
Q22 / choice	100	100 / 100	238–245	34	—	—
Fast / chain	100	100 / 100	333–343	34	61–63	0
Fast / choice	100	100 / 100	240–247	34	49–50	0

The experimental fast profile adds a distinct 66-byte application frame: the fourteen-byte packet identity, a sixteen-byte source–sink HMAC, the twenty-byte hop header and its sixteen-byte hop tag. The sink emits this private receipt after verifying source data, then generates the original Ed25519 proof. Relays check the hop envelope and pending route and forward the private receipt opaquely; they assign no forwarding credit from it. Only the source authenticates the end-to-end MAC and settles its service debt early. The separate public proof remains subject to Ed25519 verification and the original MAC-ACK credit gate. A private MAC is not transferable public evidence, and its deadline endpoint is not the original signed-receipt endpoint.

Fixed-point inference reduces signed-receipt misses from 134/200 to 68/200; the 90 ms class still fails. In the fast profile, all two hundred private receipts meet their source deadlines, while all two hundred deferred Ed25519 proofs are eventually verified and sixty-eight miss the original signed deadline. Extra private receipts add three radio frames per chain packet or two per choice packet; no energy saving is inferred. Tested-path stack watermarks reach 6624 bytes, with 3968 static RAM bytes and intact guards under the 32 KiB linker budget. These are observed paths, not a worst-case stack bound.

A separately frozen compiler-only O3 pilot repeats one hundred packets in each topology with the same Q22 actor, signature-verification order and original deadlines. Both runs deliver all offered packets and source-verified Ed25519 receipts, and all five hundred decisions match the frozen reference. Signed delays are 331–344 ms in the chain and 239–247 ms in the choice geometry; each run still misses thirty-four deadlines. Seven authenticated negative controls pass per run. The Linux Renode run and historical Mac O2 run are descriptive diagnostics, not paired statistical replicates.

All sixty-eight misses are in the 90 ms critical class; neither of the other classes misses its deadline. UART stage records show median sink signing of 30 ms and median signature verification of 82 ms at the source and forwarding relays; per-call HMAC medians are 1 ms. These cryptographic stages dominate the emulated receipt path. The stage records include control operations and are descriptive emulator timings, not independent packet replicates or physical measurements. Compiler optimization therefore does not resolve the original deadline requirement. Public rechecking verifies all two hundred sink-generated signatures; private hop-tag rechecking requires the intentionally unexported ephemeral fixture keys.

Fresh pairwise keys and a signing seed are provisioned once over the trusted emulator UART; persisted host allocation reserves distinct boot sessions. The software rejects stale sessions, repeated provisioning, mismatched signing keys and forged fast MACs, and uses explicit wipes for temporary key material and UART buffers. Source replay state rolls across longer streams. None of this establishes protected MCU storage, authenticated physical provisioning or reboot-persistent device state. The pilot still fixes four nodes, nominal energy/queue inputs and test-triggered beacons. Physical RF/energy, overload, autonomous rendezvous and deployment behavior remain unvalidated; retrospective private hop-tag audits require the locally retained fixtures.

5.9 Cooja transfer and relay behavior change

A separate Cooja evaluation [25] runs the frozen Q22 actor and weighted feedback ledger inside four native Contiki-NG motes. Thirty new simulation seeds cross five arms, benign or post-warm-up blackhole behavior, and native or scheduled processing time: 600 runs and 36,000 offered packets. The source offers sixty packets at 300 ms intervals; two feasible relays connect it to the sink. All arms share geometry, traffic, authenticated beacons, nominal energy/queue inputs and the same checkpoint. Censored-plus-proof retains the censored evidence rule while enabling the active arm’s proof-generation rule; realized traffic can still differ as routes change. This is a four-node policy-transfer experiment, not a repetition of the original network-scale benchmark.

Actual CSMA callbacks supply first-hop MAC outcomes. A radio-driver hook observes successfully received forwarding frames before destination filtering, rather than reading simulator-held forwarding histories. Passive evidence has strength 0.5 and a signed sink receipt strength 1; the Python ledger independently replays ACK gating, 150 ms expiry, maximum-strength deduplication and late upgrades. Active proofs are authenticated hop messages emitted after successful downstream MAC outcomes. Public fixture keys support trace verification, not deployment secrecy. Sixteen local pending routes expire after two seconds; closed feedback records can remain eligible for late upgrades until their slots are reused.

Native execution does not account for Cortex-M3 instruction time. The delayed setting schedules outbound data after 14 ms, other outbound messages after 1 ms, received public receipts after 81 ms, sink data processing after 31 ms and other received frames after 1 ms. These per-packet assumptions do not model serial CPU service, overload or measured compute energy. Table 11 reports means over thirty seeds; Fig. 12 shows paired delivery contrasts, with a shared horizontal scale within each attack setting. Pointwise bootstrap intervals use 20,000 seed resamples. Two-sided paired sign-flip tests use 100,000 draws, with Holm adjustment across sixteen contrasts within each endpoint family. Delivery is primary; critical sink-deadline failures and total transmit-state duration form separate secondary families. Four direct matched-proof contrasts per endpoint were added after the protocol freeze as exploratory checks, with a separate Holm family; they do not alter primary inference.

Table 11. Cooja transfer results.

Setting	Feedback arm	Delivery base (%)	Delivery delayed (%)	Critical failure delayed (%)	TX delayed (s)
Benign	Active proofs	98.78	99.17	94.17	8.282
Benign	Naive passive	98.67	98.39	95.17	7.933
Benign	Frozen prior	98.33	98.50	95.00	7.932
Benign	Censored + proofs	98.50	98.72	94.50	8.277
Benign	Censored passive	98.33	98.50	95.00	7.932
Blackhole	Active proofs	71.39	79.11	96.00	7.931
Blackhole	Naive passive	47.00	54.89	96.67	7.341
Blackhole	Frozen prior	32.83	32.94	98.17	7.048
Blackhole	Censored + proofs	33.17	32.83	97.50	7.169
Blackhole	Censored passive	32.83	32.94	98.17	7.048

Delivery counts unique sink packets; critical failure includes missing critical packets or the sink application delivery event later than 90 ms. TX is the sum of modeled transmit-state seconds over all four motes, not energy. Base uses no added instruction-cost delay; the delayed column uses the declared processing assumptions. Full per-arm intervals, proof counts and adjusted contrasts are retained with the raw traces.

With scheduled processing and a blackhole relay, active proofs change delivery relative to censored feedback by +46.17 percentage points (95% paired interval +36.89 to +54.50; Holm-adjusted p=0.00016). Enabling proofs while retaining censored evidence changes delivery by -0.11 percentage points (95% paired interval -0.50 to +0.39; Holm-adjusted p=1). These are conditional transfer results; they neither replace the original Holm family nor establish a new neural-policy advantage. Across all captures, independent reconstruction differs by at most 1.19e-07 per feature, with maximum logit discrepancy 2.21e-06.

The four-node geometry, one source, fixed nominal energy/queue/LQI telemetry, public test keys and unicast beacon roster limit generalization. The fixed-point port quantizes float timestamps to integer

milliseconds; a beacon can be excluded about 1 ms earlier than its ideal acquisition timestamp. The corrected auditor reproduces that representation without changing captures or outcomes. Energest state durations do not establish current or battery consumption, and MAC ACKs are not newly cryptographically authenticated. Censoring preserves earlier positive evidence when later observations are absent; recovery from a previously cooperative relay that becomes silent therefore depends on other routing inputs or additional evidence.

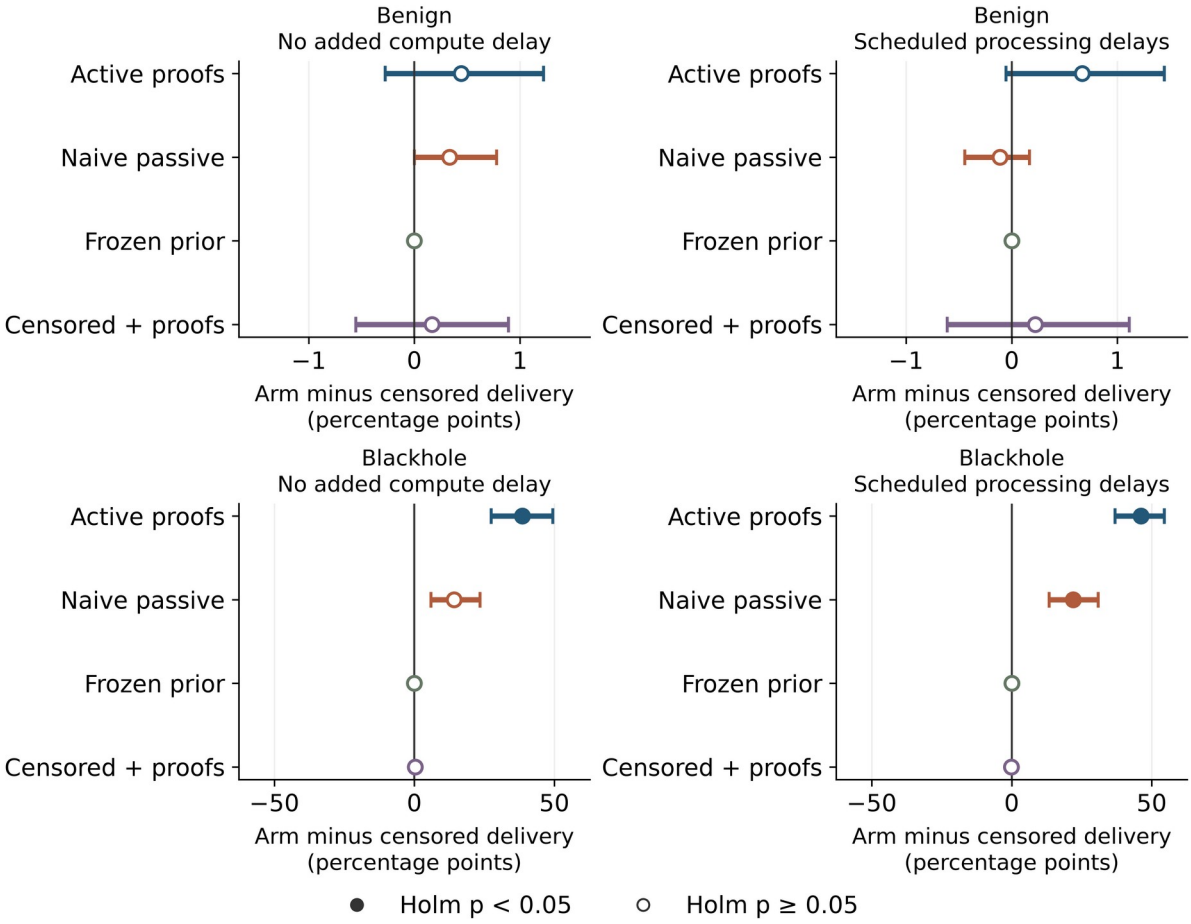


Fig. 12. Paired Cooja delivery effects.

5.10 Archive diagnostics

The archived censored and frozen-prior captures are compared post hoc by simulation seed, attack/timing stratum, observer and packet identity. All 11,927 decisions across 120 paired runs match; no decision is unmatched. Risk coordinates and recorded valid logits differ in 7,023 decisions, while non-risk coordinates, candidate ordering and selected relays remain identical. The maximum paired valid-logit difference is 0.15597. Of 7,107 multi-choice decisions, 120 have exact top-logit ties in each arm; action equality is therefore not merely a consequence of single-candidate masks. Effective-count trajectories preserve all 143,928 archived feedback events. Figure 13 separates these descriptive counts from deadline outcomes. Equality of delivery is therefore accompanied by equality of recorded actions in this archive, rather than being used to infer it.

A deterministic grid intervenes on one candidate's two risk coordinates at a time in 120 evenly spaced multi-choice source observations per attack/timing stratum. The five effective-count pairs are (1,1), (5,1), (20,1), (1,5) and (1,20); the retained candidate set and other coordinates are held fixed. The

compiled Q22 actor changes its action in 1,010 of 4,800 interventions, with maximum logit change 0.18371. These are offline actor-sensitivity checks, not closed-loop delivery results, independent replicates or a test of candidate preselection. They show that the observed positive-count changes were insufficient to change choices, not that every risk input is ignored.

For benign censored runs with scheduled delays, 600 critical packets include thirty timely deliveries, 563 delivered-but-late packets and seven undelivered packets. The median sink application delay is 131 ms and source signed-receipt delay 382 ms over their respective received packets across all classes. Thus, most critical failures are lateness rather than loss. Sink arrival, application processing and reverse receipt timing remain distinct endpoints; high delivery does not establish the original 90 ms service target. Antecedent-event checks pass on all 600 archived runs, binding emitted feedback to prior received frames, processing readiness, own MAC outcomes and packet identities. They complement wire/ledger/policy audits rather than proving an independent RF model.

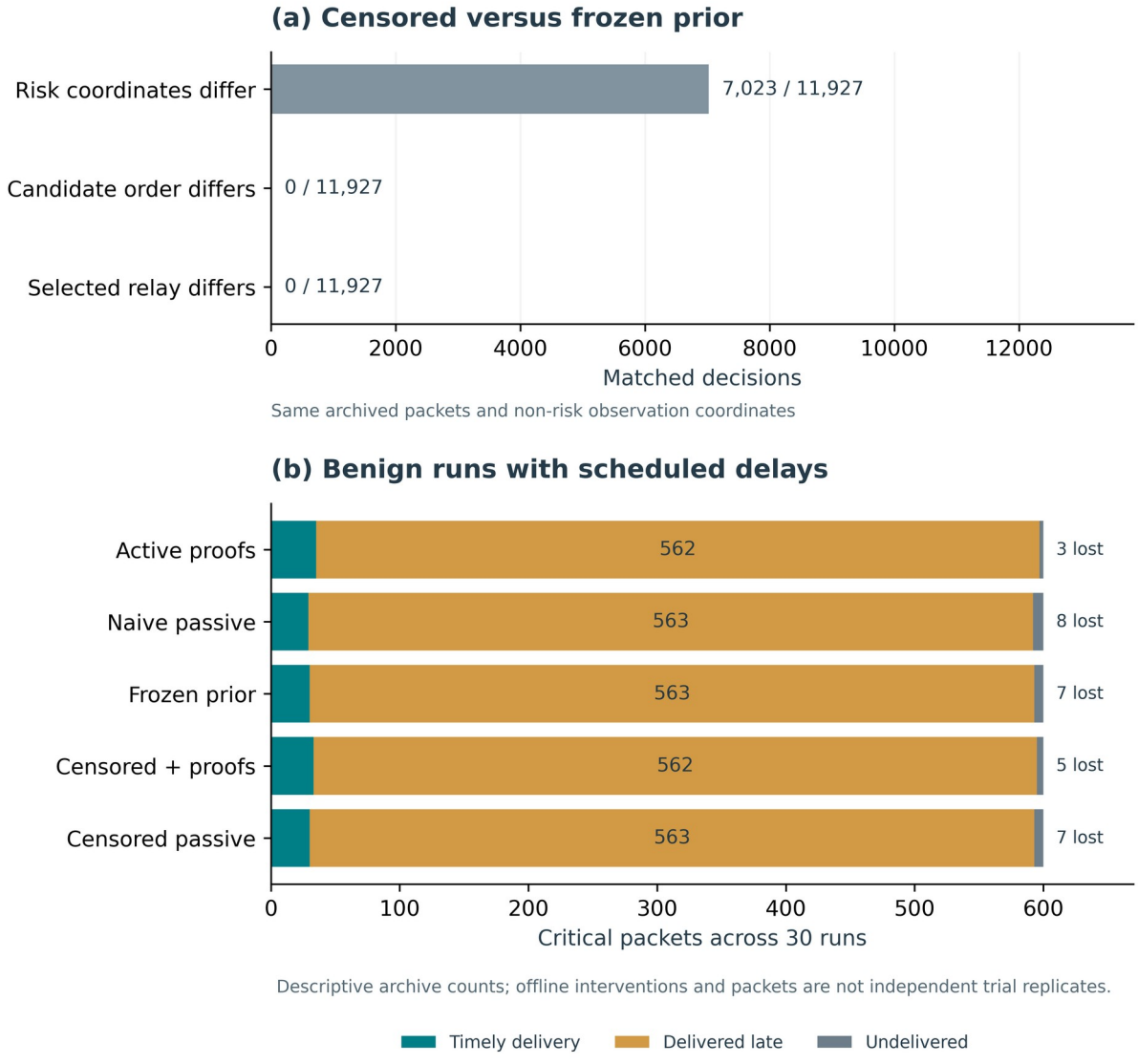


Fig. 13. Feedback and deadline diagnostics.

5.11 Matched feedback and counterbalanced transfer

The new protocol was committed before new radio outcomes as b24418008b318aa4a082e0a666c4a2196b5c994c; pre-primary-outcome amendment

40a933ee06b0dd1521f7de36d4b215f9f8d8b9d4 records GCC portability corrections after an excluded compile-only preflight failure. Twelve independently perturbed four-node geometries use three retained checkpoints. Each checkpoint covers onset packet 10 or 30 crossed with UDGM RX probability 0.9 or 1. Two radio seeds per block cross both relay-ID permutations, benign behavior or either physical attacker position, both timing settings and five arms: 1,440 runs. Perturbations are uniform within ±1 m around each original relay position. Source and sink remain fixed; the geometry family is narrow and the block factors are not fully crossed within each geometry. Relay-ID permutations also exchange the inherited 50 ms beacon-phase offsets and identity tie-breaking order. Both permutations are paired across arms rather than treated as interchangeable radio histories.

Four arms cross a missing-evidence failure increment with proof generation. Algorithm 1 and Fig. 2 retain the censored lifecycle; the negative-update arms add the declared failure increment at expiry. Figures 1 and 7 continue to describe their original simulation and host-transport stages, not a production deployment. All four require the observer's own ACK, expire 150 ms after it, apply the same incremental clipped positive weight and admit late positive evidence until slot reuse. Missing-evidence penalties use the cached expected-link weight once at expiry. A late positive observation adds positive evidence without retracting that earlier penalty. Application offers and processing rules are fixed. Enabling proofs intentionally changes control airtime and the associated scheduled processing; these costs are part of that intervention. Proofs do not enter these four ledgers, and route-dependent realized proof counts can differ. The fifth arm retains legacy active-proof eligibility and expiry as a separate reference, with the shared stricter proof source/session/pending-route check. Public fixture keys exercise serialization and validation, not deployment secrecy. Native execution and overlapping per-packet scheduled delays retain the scope of Section 5.9.

The Cooja port uses 38 m to normalize sinkward progress in preselection and candidate features, whereas Eq. (1) describes the original 50 m benchmark normalization. Candidate progress is clipped to $[-1, 1]$; both source relay coordinates saturate at 1 throughout this narrow fixture family. Preselection still uses unclipped progress, and a geometry perturbation can admit an additional positive-progress relay-to-relay choice. Only sinkward candidates are retained. These inherited transfer rules, nominal telemetry and fixed sources are common across the matched arms; the experiment does not restore the original observation distribution or establish wide network-scale generalization.

The remediation questions were motivated by the archived failure; they are exploratory relative to the original study, with their own analysis prespecified before the new runs. Primary inference within this new family uses post-onset unique sink delivery in attacked runs. Each geometry block averages the two radio seeds, two identity permutations and two physical attacker positions within timing and arm. Negative-update and proof-generation main effects average their two corresponding simple effects; the interaction is the difference in differences. Pointwise 95% intervals use 20,000 paired geometry-block bootstrap resamples. Two-sided exhaustive sign-flip tests enumerate 4,096 sign assignments over twelve block differences, conditional on symmetric block effects; this is not a claim of randomized treatment allocation. Holm adjustment covers six primary contrasts, three per timing setting. Whole-run critical failures and total transmit-state seconds have separate six-contrast secondary families. No equivalence or noninferiority claim is made. Table 12 and Fig. 14 report the primary effects.

Table 12. Matched delivery effects.

Timing	Contrast	Effect (pp)	95% interval (pp)	Holm p
Native	Negative update	+19.000	[+11.073, +27.198]	0.005859
Native	Proof generation	+0.597	[-0.642, +1.865]	0.668
Native	Interaction	+2.569	[-0.028, +5.167]	0.3555
Scheduled	Negative update	+21.170	[+14.410, +27.834]	0.005859
Scheduled	Proof generation	+1.017	[-0.826, +2.906]	0.668
Scheduled	Interaction	+2.563	[-0.806, +6.271]	0.6548

2 of the six primary contrasts survive Holm correction: native negative update, scheduled negative update. The remaining contrasts do not establish equivalence or the absence of a practically relevant effect.

Effects use percentage points and equal geometry-block weight; packets are not independent replicates. Native means no added compute delay, not calibrated zero MCU processing cost. The intervention estimates include downstream routing and traffic consequences. The active reference is not part of the matched factorial family, and its additional comparisons remain exploratory. All source captures, compressed native images, case settings and checkpoint identities are rebound by hash before this manuscript integration; historical statistical families remain unchanged. Per-case audit acceptance denotes complete and consistent reconstruction, not perfect delivery or satisfaction of a service deadline; loss and lateness remain measured outcomes.

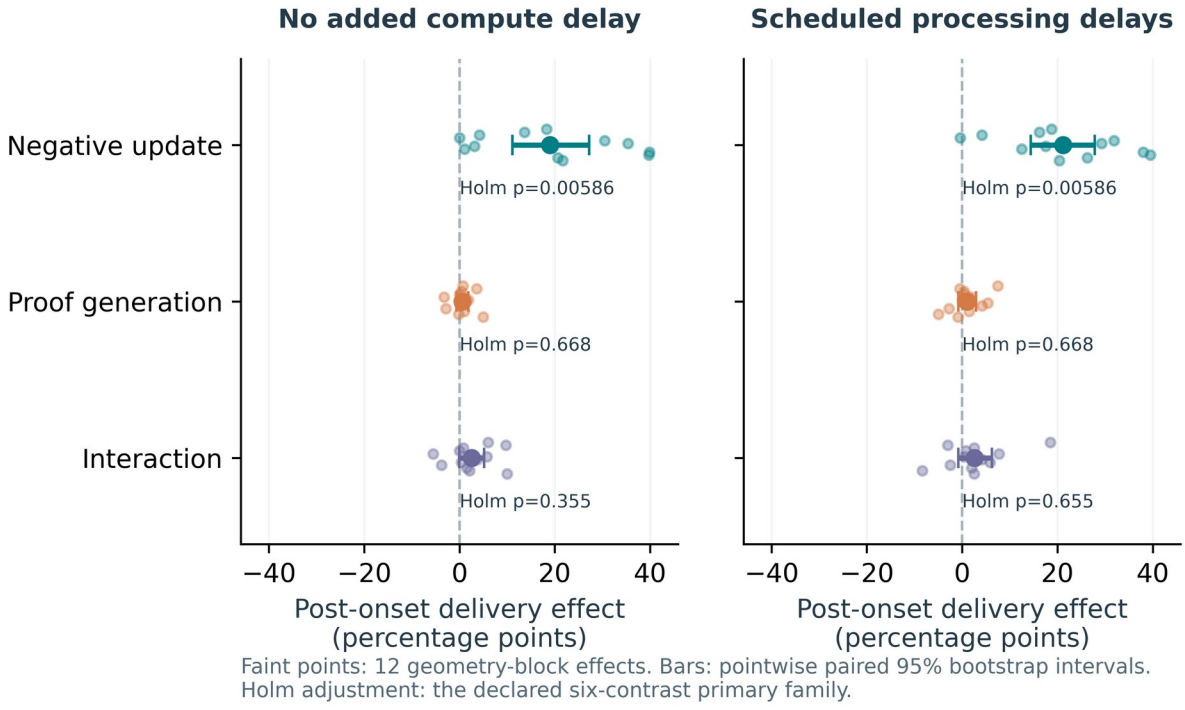


Fig. 14. Matched feedback effects.

Under scheduled processing, the numbers of attacked run settings meeting the fixed twenty-opportunity recovery criterion are censored 48/96; censored with proofs 48/96; negative update 68/96; negative update with proofs 72/96; active reference 69/96. Right-censored settings over the full available window are censored 48/96; censored with proofs 48/96; negative update 15/96; negative update with proofs 13/96; active reference 19/96. These are descriptive tallies over correlated seed, permutation and attacker settings, not additional independent experimental units. The retained secondary summary reports conditional confirmation latencies alongside censoring, cumulative post-

onset losses, relay switches and proof frames/bytes. Feedback trajectories preserve effective counts and their heuristic risk coordinates; no observed recovery fraction is used to tune the policy or select an onset threshold.

An additional post-outcome exploratory aggregation conditions on selecting the attacker in the final pre-onset source decision. Within twenty opportunities, the counts are censored 0/48; censored with proofs 1/48; negative update 20/48; negative update with proofs 25/48; active reference 22/48. All forty-eight censored settings in this subgroup remain right-censored over the full remaining window. Among observed subgroup recoveries, median confirmation times are 4.8 s for negative updates, 4.5 s for negative updates with proofs and 4.5 s for the active reference; these conditional medians exclude fifteen, twelve and eighteen right-censored settings, respectively. They are descriptive, do not alter the primary family and do not establish safe downstream paths. Proof-enabled matched arms average 38.90 or 48.89 proof frames per attacked scheduled run under censored or negative updates, respectively; realized counts depend on routing.

Adaptation is assessed as the first three consecutive source choices avoiding the declared attacker, with right censoring when no such sequence occurs. Recovery within twenty packet opportunities uses the same fixed observation window for both onset settings; a single safe choice is not sustained recovery. The criterion includes source choices already avoiding the attacker at onset; meeting it does not by itself establish adaptation. Additional exploratory summaries stratify by the last pre-onset source choice and count post-onset switches. This is a routing-choice diagnostic rather than proof of subsequent delivery. The budget fixes twelve geometry blocks; it is not a power justification for declaring small effects absent. The small balanced fixture family, shared four-node graph, fixed source/sink, three checkpoints, public keys and nominal energy/queue telemetry limit transfer. Hop-authenticated relay claims also do not establish truthful forwarding by a malicious key holder, and the runtime router cannot access the independent auditor’s global trace.

5.12 Compiler optimization and radio current sensitivity

A further compiler diagnostic was frozen at commit b52842496dc625ad30ff95bdefad4c107366080e before execution in workflow 39179074994. Link-time optimization (LTO) and default loop optimization supplement O3; the Q22 actor, Ed25519 endpoint, clock model, traffic and 90/350/1500 ms deadlines remain unchanged. All 200 offered packets produce source-verified signed receipts, all 500 routing decisions match the frozen reference, and seven authenticated negative-control categories pass in each topology. Table 13 compares the retained diagnostics. These sequential runs use fresh private keys and are not independent statistical replicates of a compiler speedup.

Table 13. Signed receipt compiler diagnostics.

Profile	Topology	Signed delay (ms)	Misses / offered	Decisions
O3	Chain	331–344	34 / 100	300
O3	Choice	239–247	34 / 100	200
O3 + LTO	Chain	299–308	34 / 100	300
O3 + LTO	Choice	213–220	34 / 100	200

The LTO profile still misses every critical-class deadline: 34/100 packets in each topology. Packet-matched sink signing plus source signature verification occupies at least 104 ms in the chain and 105

ms in the choice geometry; both medians are 106 ms. These two serial dependencies alone exceed 90 ms before relay verification, routing and radio overhead are added. The observations describe this emulator and implementation; they are not universal lower bounds on Ed25519 or physical-device timing. Independent public-key checks verify all 200 sink signatures, match all 200 accepted source receipts to received signed frames, and reject 200 one-bit signature mutations. Private hop tags cannot be retrospectively reverified without the intentionally unexported per-run secrets.

A separate exploratory reanalysis checks all 600 retained historical Cooja runs: thirty seeds per attack, timing and feedback-arm stratum, 2,400 final node records and 36,000 offered packets. Every decompressed capture matches its archived hash; transmit/listen state accounting and source/sink endpoint counts match the retained audits. No new simulations are represented by this reanalysis. Fig. 15 relates descriptive delivery means to summed node transmit time. The original active and censored arms retain their historical semantic differences; this plot is not a substitute for the matched-factorial inference in Section 5.11.

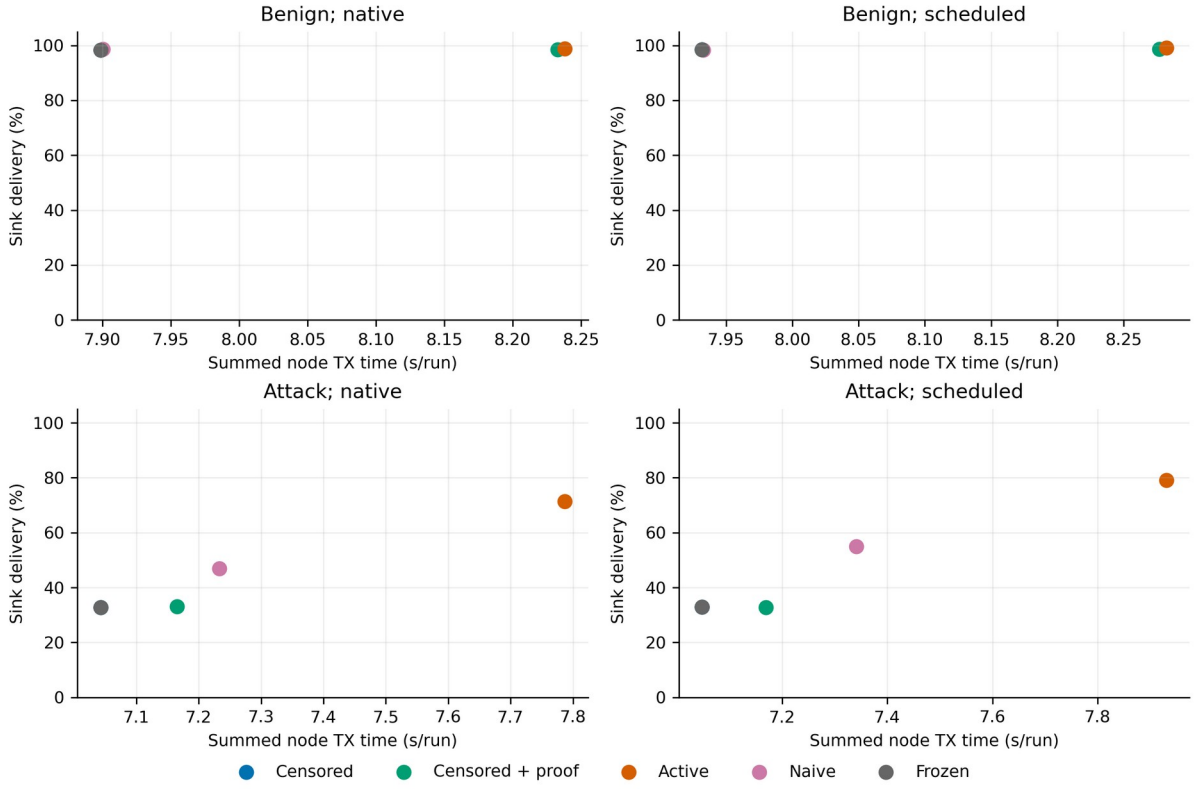


Fig. 15. Delivery and transmit airtime.

For a common supply voltage V (volts) and constant listen current I_L (amperes) across nodes, the radio-only model is $E_{radio} = V\, I_L\, (T_L + r\, T_{TX})$, where T_L and T_{TX} are summed node state durations in seconds and $r = I_{TX}/I_L$ is dimensionless. Fig. 16 varies the assumed ratio over 0.5, 1, 2 and 3. The plotted relative values require no assumed absolute current or voltage. At r = 1, every arm has equal radio energy over the always-on accounting window; rankings can reverse across this point. CPU, cryptography, sensing, regulator and host consumption are excluded. These are conditional model calculations, not measured currents, joules or battery lifetimes. Coincident plot markers indicate equal means.

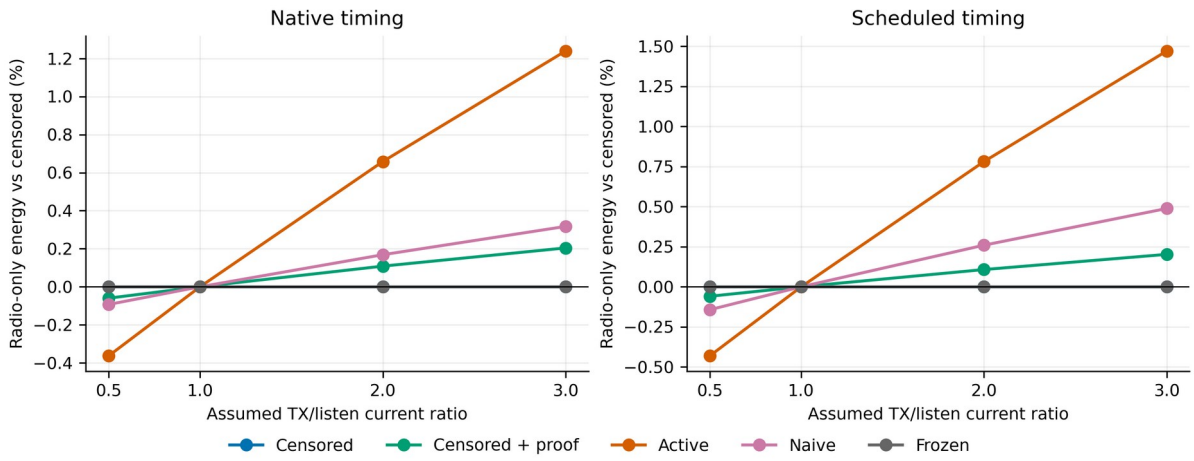


Fig. 16. Assumed radio current sensitivity.

The retained datasets therefore support virtual radio-state and endpoint checks. Calibrated physical RF/current acquisition, board-specific protected keys, authenticated provisioning, firmware authenticity and reset-safe persistent MCU sessions remain unvalidated. The native Cooja timing assumptions do not supply a serial CPU model or physical calibration.

6. Discussion

6.1 Critical vulnerabilities and limitations

The evidence supports a conditional feedback comparison. Censored feedback meets the original 750-run decision rule, yet loses substantially to active proofs in the separate delayed Cooja blackhole setting. These studies differ in topology, traffic, state construction and execution, so their contrast does not identify behavior change as the sole cause. Within Cooja, the censored-plus-proof control tests whether enabling proof generation under the retained censored rule restores delivery; it does not. Eligibility and expiry semantics still differ between active and censored arms, preventing attribution of the full gap to one update rule. Section 5.11 removes the stated ACK/expiry/late-positive differences within four new matched arms. Its factorial effects apply to that intervention and fixture family; they do not retroactively identify the complete cause of the original 750-run gain.

The frozen benchmark relies on simulator-held geometry, packet history and reverse receipt paths; beacon values are supplied through receive-gated metadata. Source-only service debts leave transit queue control untested. Its threat model also excludes forged receipts, deceptive telemetry, compromised sources, collusion and mobility. Overheard transmission confirms an attempt, while successful delivery at the next hop remains uncertain. The effective-count risk score should be read as a routing preference, not an evaluated live intrusion detector.

A four-second horizon, a central sink and a 60-packet burst are insufficient to establish steady-state scalability or network lifetime. After cache sorting, candidate scoring is linear in the bounded candidate count. Record expiry is more costly: each millisecond poll scans all W retained watches, giving O(W) work and O(W) memory. The frozen experiments provide neither a long-horizon reclamation policy nor measured embedded inference cost. Sections 5.6–5.7 add bounded-state tests and isolated instruction-level execution; physical integrated-firmware costs remain unmeasured. An always-listening radio also omits the wake-up cost that evidence collection would incur under duty cycling.

The 150 ms evidence timeout includes contention, retries and forwarding. Interference can therefore delay a valid receipt past expiry. Censoring avoids assigning failure solely from silence and allows a

later receipt to upgrade the record; it does not establish an optimal timeout. The simulator also uses a shared time base. The corrected host timestamp sampling error is different from physical clock offset or drift, which remain unmeasured; Section 5.7 tests a declared virtual envelope.

The holdout fixes passive strength at 0.5, record expiry at 150 ms, cache life at 0.75 s, the current model and all inference checkpoints. Section 5.5 varies the preselection coefficients only; sensitivity to these other choices remains unknown. Changes to serialization, ACK authentication, service tracking or ledger retirement require a separately frozen experiment because they change the evaluated mechanism.

6.2 Interpretation of the exposed failure boundary

With the rate-anomaly term disabled, censored feedback keeps the effective failure count β at 1 and never decreases the effective success count α. If no new accepted evidence arrives, these counts and their derived risk coordinates remain unchanged. Positive observations can increase α, but subsequent silence supplies no corrective count. This follows from Eqs. (3)–(5); it is a property of the declared estimator, not evidence that every learned routing action must remain fixed. Cache freshness, link estimates, geometry and service debt can still change the selected relay.

Table 11 and Fig. 12 expose the practical consequence in the tested four-node transfer setting. The delayed active-minus-censored delivery contrast is +46.17 percentage points (95% paired interval +36.89 to +54.50; Holm-adjusted p=0.00016). Adding proof generation to censored feedback changes delivery by −0.11 percentage points (95% paired interval −0.50 to +0.39; Holm-adjusted p=1). The second contrast supplies no evidence of recovery from proof generation alone; its nonsignificance is not a population-equivalence result. Active-proof semantics include different eligibility and expiry rules, so the contrast does not isolate the effect of negative updates.

The historical Cooja study exposes a failure condition, while the remediation study varies onset, channel condition, geometry and retained checkpoint within a declared small fixture family. Two onset settings and three checkpoints do not identify a universal change-point or a safe passive-feedback operating region. The balanced block design also does not fully cross every geometry with every checkpoint/onset/channel setting. An optimal probing policy remains a prospective mechanism requiring its own frozen evaluation.

6.3 Future experiments

Section 5.6 separates proof generation from the declared evidence rule, but its periodic sleep primitive lacks rendezvous. An observation policy could choose silence, passive listening or an authenticated probe according to the expected reduction in delivery loss relative to measured radio and processing cost. A discriminating deployment experiment would integrate rendezvous, test adversarial receipts and clock uncertainty, and reserve new physical-board settings. This would evaluate evidence acquisition under execution costs not established by isolated or small-scale integrated emulation.

7. Conclusions

Matched feedback controls now separate the missing-evidence update rule from proof generation in 1,440 Cooja runs over twelve perturbed-geometry blocks. In the scheduled setting, their post-onset delivery effects are +21.170 percentage points (95% pointwise interval +14.410 to +27.834; Holm-adjusted p=0.005859) for negative updates and +1.017 percentage points (95% pointwise interval -0.826 to +2.906; Holm-adjusted p=0.668) for proof generation. These estimates complement the retained conditional 4.67-point gain and later transfer failure; they do not establish a universally

preferable mechanism. Historical policy comparisons supply no established attention advantage, and the remediation experiment does not retest architectural superiority.

Exploratory reconstruction explains the censored–frozen equality in the archived setting: recorded risk changes did not alter selected relays, although larger offline risk interventions could. Failure under the tested conditions therefore cannot be equated with complete actor insensitivity. Both O3 and LTO compiler profiles retain 68/200 signed-receipt deadline misses. The radio-current sensitivity in Section 5.12 also shows why lower transmit airtime alone cannot establish measured energy savings. The study supplies neither a universal switching threshold nor a production hardware validation. Physical RF, measured energy, protected device keys and reboot-persistent MCU state remain unvalidated.

Author contributions

Shivam Bhardwaj, Abhishek Kumar Pandey and Abhinandan Tripathi contributed equally to this work.

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Data and code availability: code, frozen protocol, checkpoints, historical negative outcomes, raw traces and figure sources are available at https://github.com/DrShivamBhardwaj/JICCE_LA_CPPO_Reproducibility. The original 750-run pre-outcome source freeze is 4027f071322cb5682f39329f656d76df3b68ff62. The complete earlier evidence manuscript is retained in publication/censored_feedback/. The new matched protocol and release are identified in Section 5.11 and outputs/remediation_v1/raw_archive.json.

Declaration of generative AI use

ChatGPT assisted manuscript drafting, code preparation, analysis scripts and programmatic figure plotting. All numerical results reported here come from the retained executable experiments and traces. Responsibility for scientific verification, interpretation and final approval rests with the authors.

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